Session 705P–
Enterprise Case Studies

Jackie Callahan
Manager, Enterprise Marketing

PalmSource
Conference 2000

Agenda

- Brian Ruggiero, Countrywide
- Dr. Ray Duncan, Cedars-Sinai Medical Center

Countrywide Goes Wireless…



***Wireless Technology:* Convergence!**

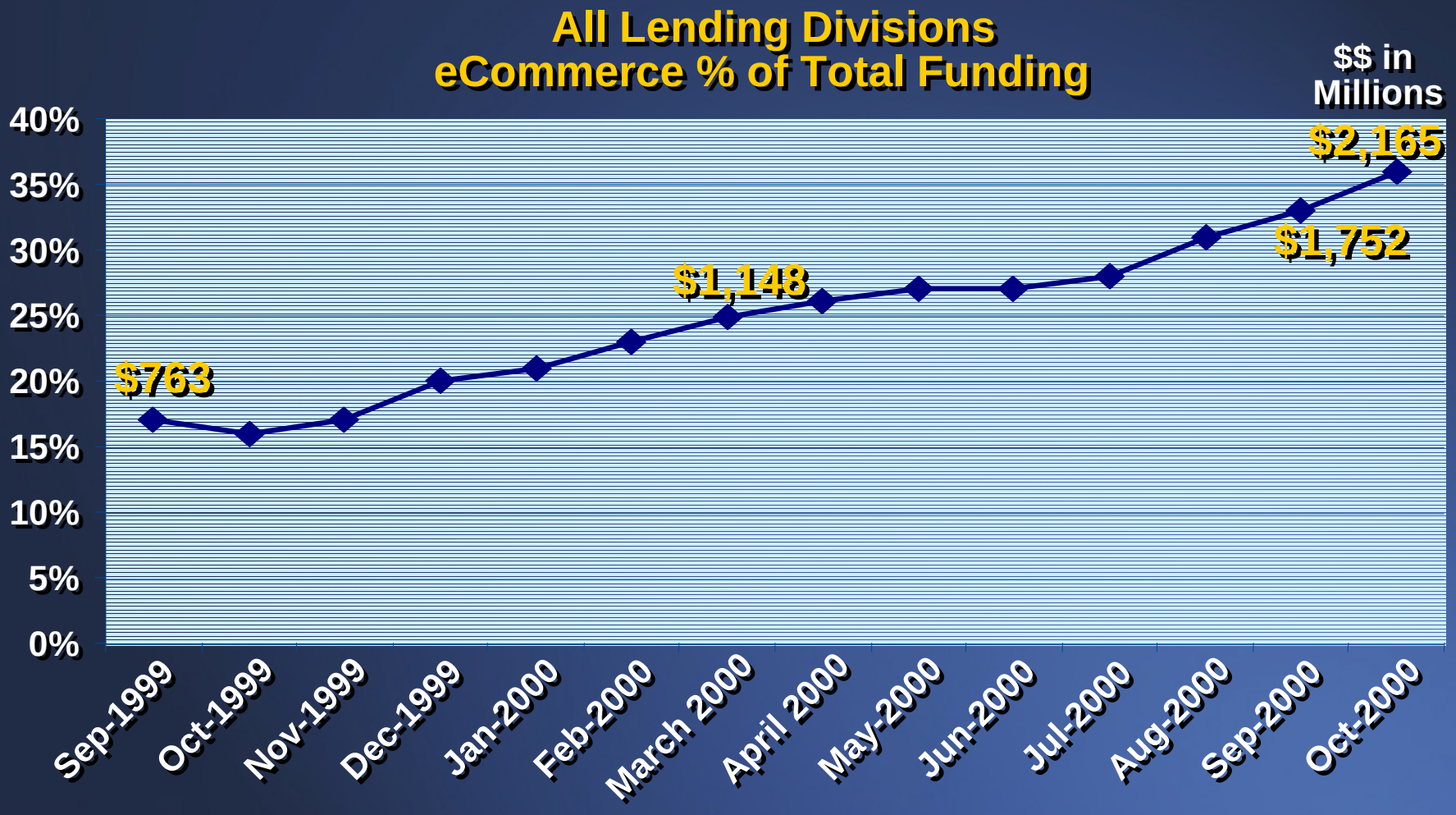
brian_ruggiero@countrywide.com

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Agenda

- Countrywide background
- Wireless background
- What Countrywide is doing
- Key principles
- Where is all this going?

The #1 Home Loan Lender on the Net…



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A Partial List of Sites…

B2C

- www.countrywide.com
- www.cwinsurance.com
- www.fullspectrumlending.com
- www.wehouseamerica.com
- https://customers.countrywide.com

B2B

- www.cwbc.com
- www.landsafe.com
- www.landsafecredit.com
- http://builders.countrywide.com
- http://realtors.countrywide.com
- https://cld.countrywide.com
  http://ccm.countrywide.com
  www.cscdealer.com

Using Highly Integrated Systems…

Origination Systems

User Services

Consumer



Brokers



BC Lending



Branch Interface



Business Services

- Marketing
- Pre-qualification
- Application
- Processing
- Underwriting
- Closing

Backend Services

- Pricing
- Artificial Intelligence
- Rules
- Documents
- Workflow
- Messaging

Interfaces

- Credit
- Appraisal
- Flood
- MI
- Banks
- Agencies

Countrywide Network

Loan Servicing

- Customer Management
- Document Management
- Investor Accounting
- Tax Services
- Foreclosure
- Real Estate Management

Data Marts

- Customer
- Application
- Property
- Account Management
- Reporting

Corporate Systems

- Accounting
- Treasury
- Secondary

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Pagers Were the Most Useful Device…

- Countrywide has traditionally had a small investment in wireless voice devices and PDAs but a large investment in paging technologies from Motorola and SkyTel



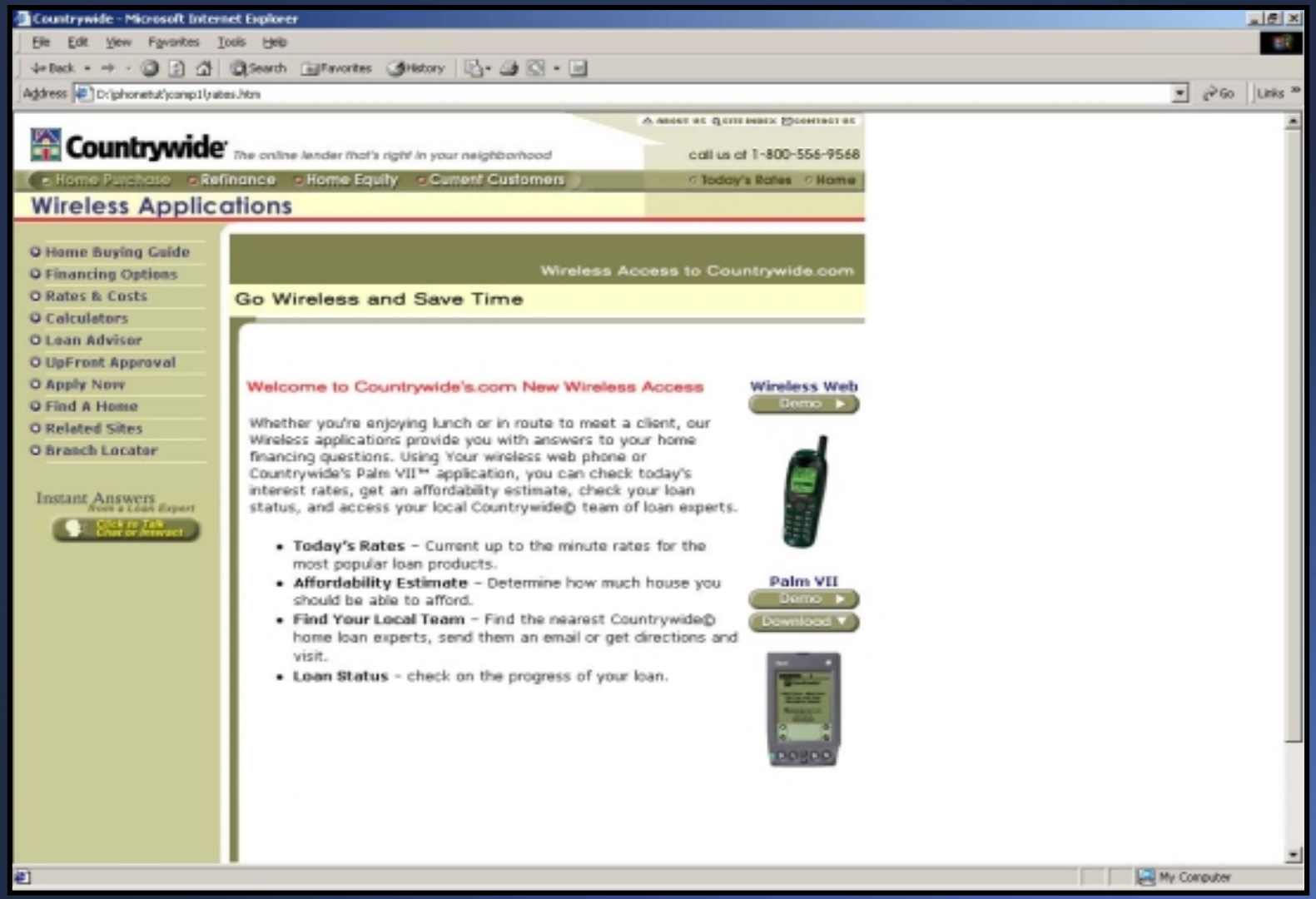
Used Very Effectively For:

- **2-way paging**
- **Integration with Countrywide eMail system (Lotus Notes)**
- **Initiate commands to support Systems**

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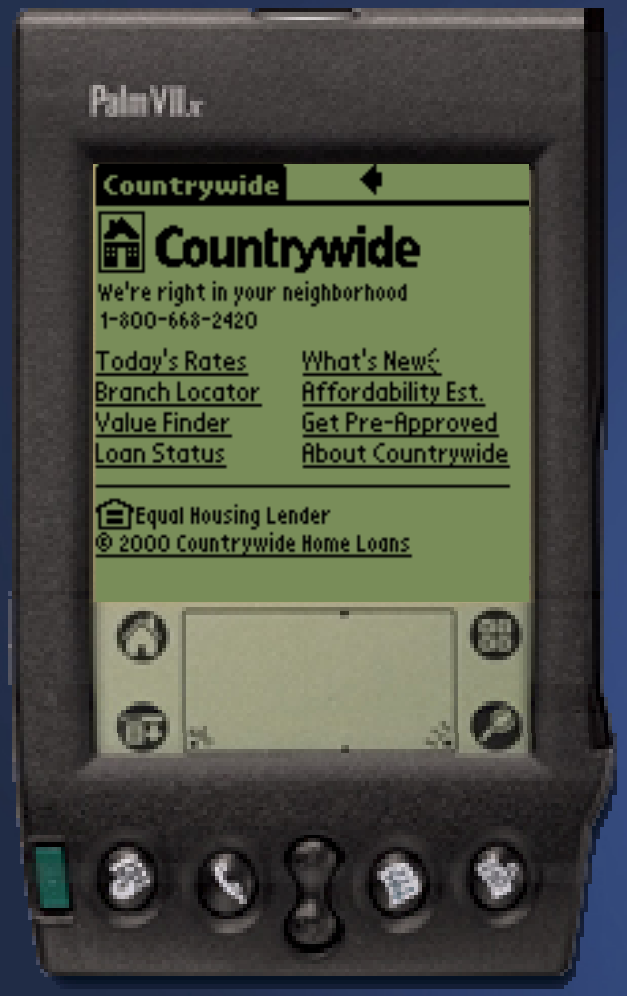
PDAs Join the Race…

- Computing power in small, handheld, personal digital assistants has accelerated to rival the power that desktop computers had only a decade ago
- The Palm™ hardware platform and Palm Operating System® have experienced wide adoption as personal digital assistants, dockable to PCs
- The Palm™ VII line and various add-on cards for earlier Palm and Palm OS OEM handhelds have integrated wireless data into PDAs, and voice capability is being achieved as mobile phone manufacturers adopt the Palm OS



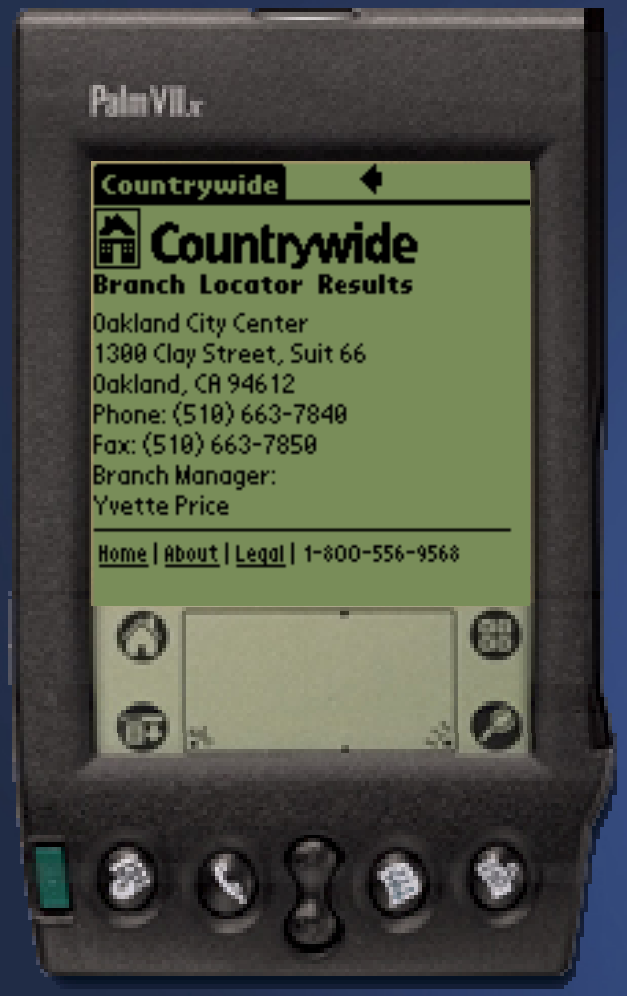
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Countrywide Palm Application…



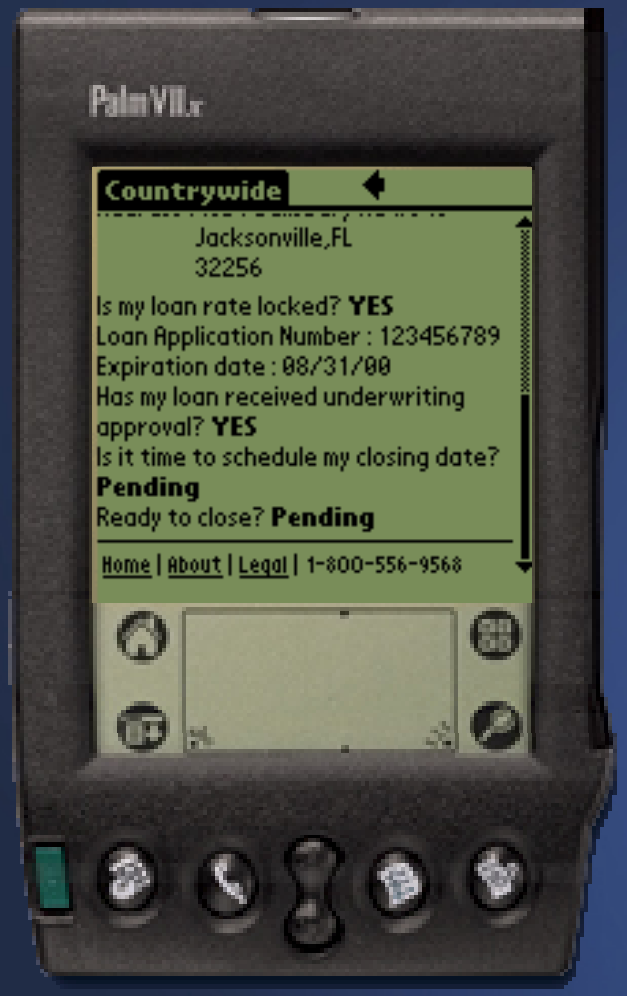
- Download the Palm™ Query Application from www.palm.net
- Palm versions of Countrywide transactions (also available on WAP phones)
- Kbrowser (WML) browser is available from 4th Pass to access WAP web sites from the Palm
- Neomar (WML and HDML) browser is in Beta

Key 1: Minimize Data Entry



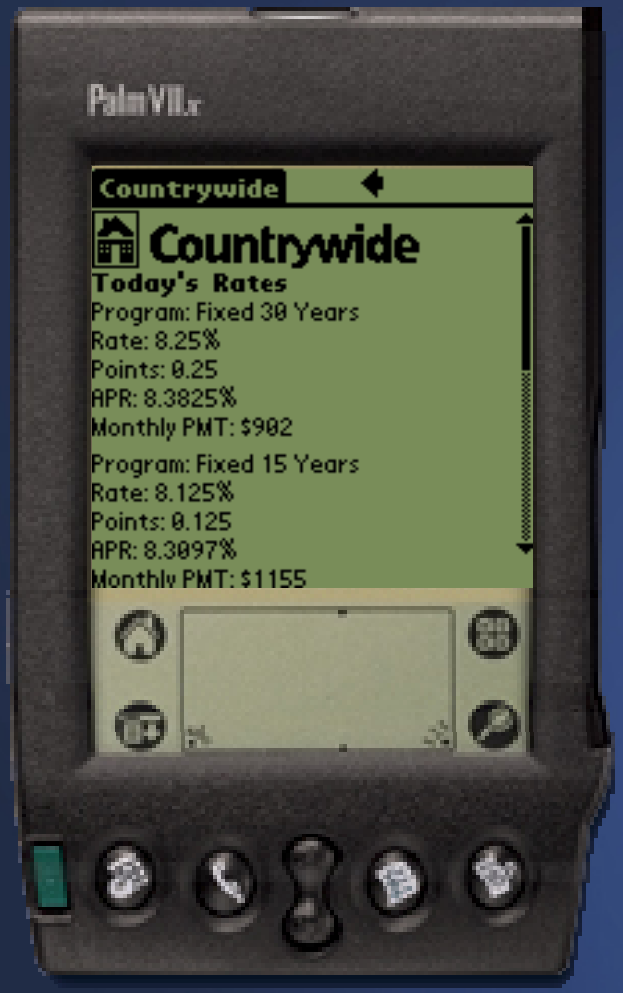
e.g., Entry of a 5 Digit Zip Code Is the Only Field Required for The Branch Locator Transaction

Key 2: Highly Useful Transactions!



e.g., Allow Countrywide Customers to Know Exactly What the Status of Their Loan Is During the Escrow Period

Key 3: Meaningful, Current Information



e.g., What Are *Today's* Rates And the Payment Amounts For All the Countrywide Products That Fit My Needs?

Convergence: Phone/PDA/Wireless Net

- Let voice take over where data leaves off! This has been useful on WAP phones, and Kyocera continues to enhance pdQ Smartphones with integrated Palm OS PDAs



Countrywide's Position

- Countrywide will continue to use technology as the key to our competitive advantage in the financial services industry
- We will excel at all channels of communications with our customers and business partners
- Countrywide sees the convergence of wireless and internet technologies as a key enabler for the next generation of eBusiness in the financial services and real estate industries

Q&A



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Enterprise Case Study: Cedars-Sinai Health System

Ray Duncan, MD
Director, Technology and Architecture
Enterprise Information Systems
Cedars-Sinai Health System

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Background



- Cedars-Sinai Medical Center is an 820 bed private nonprofit tertiary-care hospital in Los Angeles, CA
- Affiliated with UCLA Medical School, also has free-standing residency and fellowship training programs
- Business unit of Cedars-Sinai Health System
- 10,000 employees, 200+ staff physicians, 250 interns, residents and fellows, 2000 private physicians
- Not just “the hospital to the stars”

Why Palm™ VII Handheld?

- Out of the box single-vendor solution for PDA with wireless networking
- Excellent user interface and battery life, lots of applications available
- PC and Mac compatible
- Easy conversion of existing web apps
- Previous PDA/wireless projects:
  - HP 200LX plus wireless modem for ICUs
  - Alphanumeric pagers for critical lab values
  - HP Jornada 680 in Pediatric clinic and inpatient ward

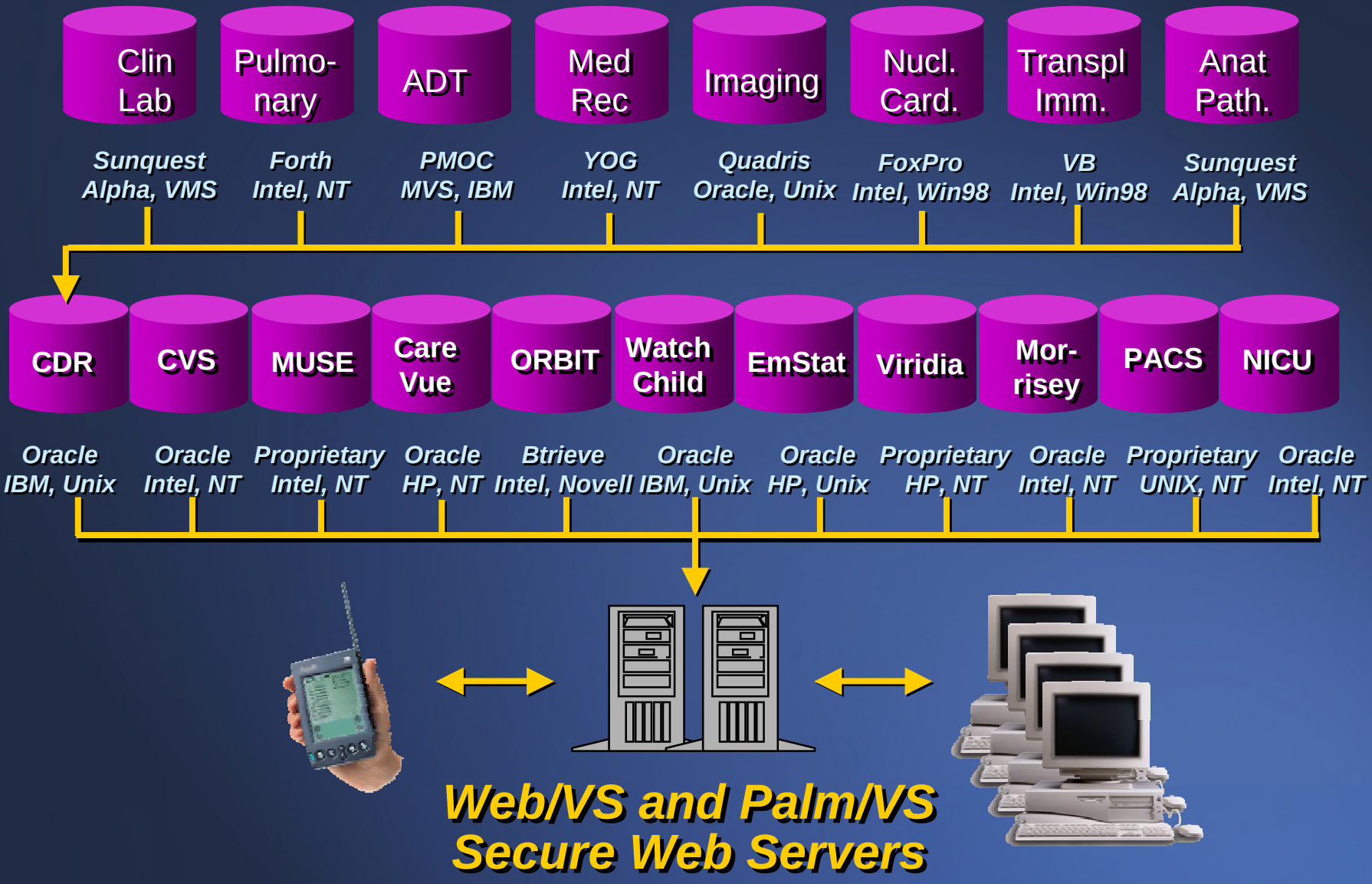


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Palm/VS—Wireless Access to Clinical Data for Physicians

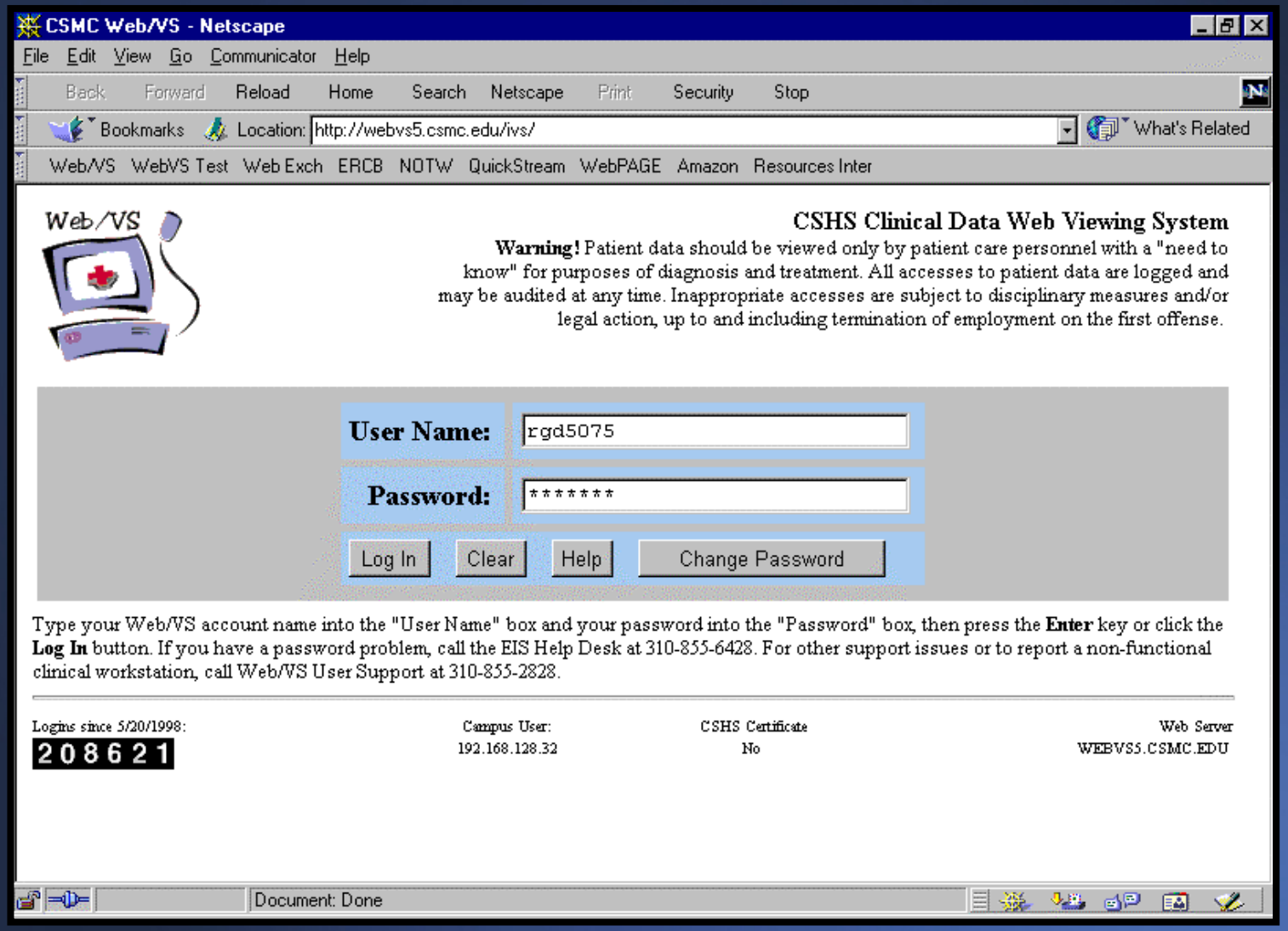
- Palm/VS is derived from a browser-based clinical viewing system (Web/VS) in use housewide since 1998
- Runs on same web servers and accesses same databases as Web/VS
- Subset of Web/VS functionality and data types
- Palm/VS is primarily used for remote access to clinical data

Palm/VS and Web/VS Architecture

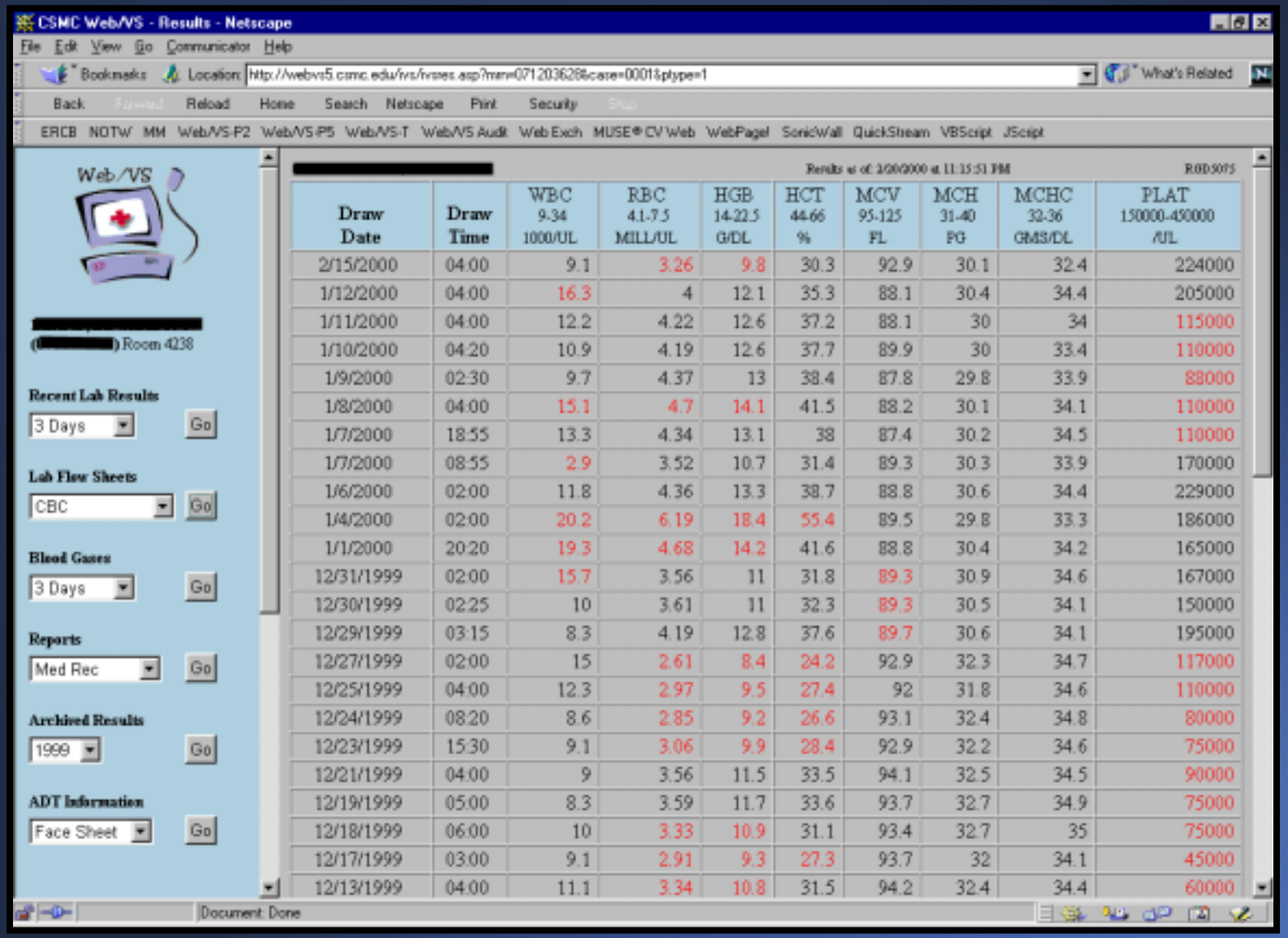


Web/VS and Palm/VS Secure Web Servers

Web/VS: Log In

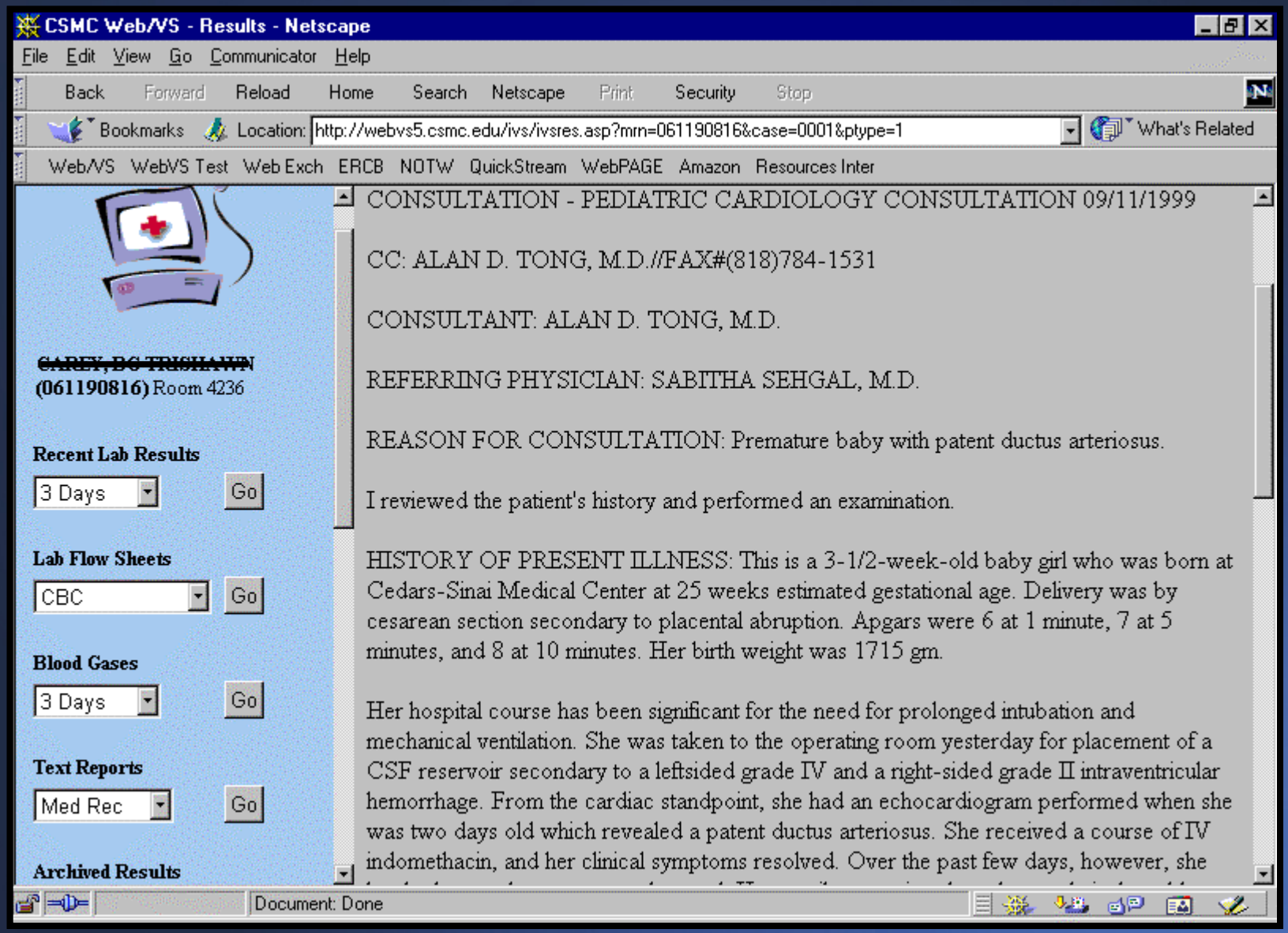

CSMC Web/VS - Netscape
File Edit View Go Communicator Help
Back Forward Reload Home Search Netscape Print Security Stop
Bookmarks Location: http://webvs5.csmc.edu/ivs/
What's Related
Web/VS WebVS Test Web Exch ERCB NOTW QuickStream WebPAGE Amazon Resources Inter
Web/VS
CSHS Clinical Data Web Viewing System
Warning! Patient data should be viewed only by patient care personnel with a "need to know" for purposes of diagnosis and treatment. All accesses to patient data are logged and may be audited at any time. Inappropriate accesses are subject to disciplinary measures and/or legal action, up to and including termination of employment on the first offense.
User Name: rgd5075
Password: *******
Log In
Clear
Help
Change Password
Type your Web/VS account name into the "User Name" box and your password into the "Password" box, then press the Enter key or click the Log In button. If you have a password problem, call the EIS Help Desk at 310-855-6428. For other support issues or to report a non-functional clinical workstation, call Web/VS User Support at 310-855-2828.
Logins since 5/20/1998:
208621
Campus User:
192.168.128.32
CSHS Certificate
No
Web Server
WEBVS5.CSMC.EDU
Document: Done

Web/VS: Flowsheet Views

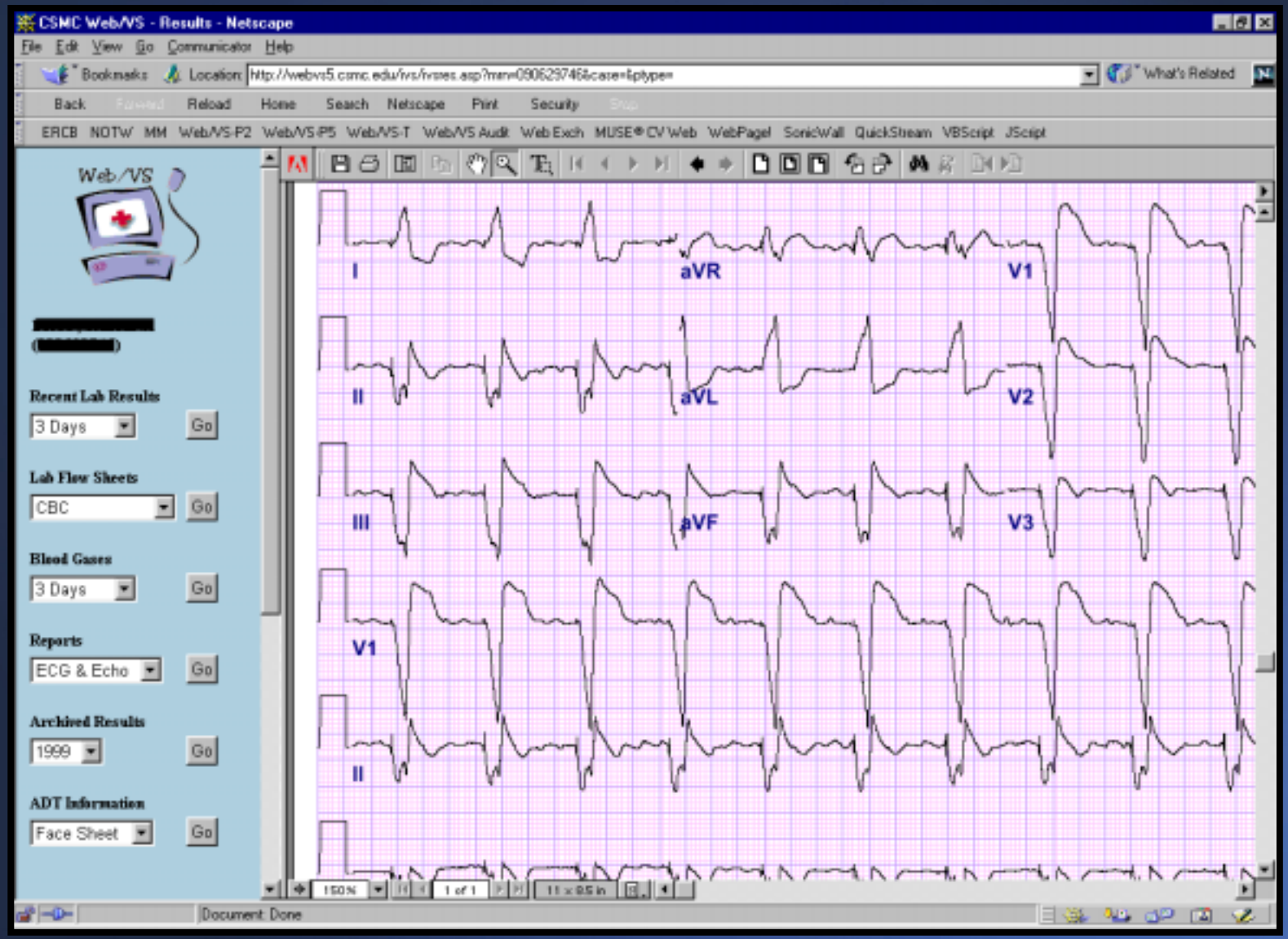


Draw Date	Draw Time	WBC 9-34 1000/UL	RBC 4.1-7.5 MILL/UL	HGB 14-22.5 G/DL	HCT 44-66 %	MCV 95-125 FL	MCH 31-40 PG	MCHC 32-36 GMS/DL	PLAT 150000-450000 /UL
2/15/2000	04:00	9.1	3.26	9.8	30.3	92.9	30.1	32.4	224000
1/12/2000	04:00	16.3	4	12.1	35.3	88.1	30.4	34.4	205000
1/11/2000	04:00	12.2	4.22	12.6	37.2	88.1	30	34	115000
1/10/2000	04:20	10.9	4.19	12.6	37.7	89.9	30	33.4	110000
1/9/2000	02:30	9.7	4.37	13	38.4	87.8	29.8	33.9	88000
1/8/2000	04:00	15.1	4.7	14.1	41.5	88.2	30.1	34.1	110000
1/7/2000	18:55	13.3	4.34	13.1	38	87.4	30.2	34.5	110000
1/7/2000	08:55	2.9	3.52	10.7	31.4	89.3	30.3	33.9	170000
1/6/2000	02:00	11.8	4.36	13.3	38.7	88.8	30.6	34.4	229000
1/4/2000	02:00	20.2	6.19	18.4	55.4	89.5	29.8	33.3	186000
1/1/2000	20:20	19.3	4.68	14.2	41.6	88.8	30.4	34.2	165000
12/31/1999	02:00	15.7	3.56	11	31.8	89.3	30.9	34.6	167000
12/30/1999	02:25	10	3.61	11	32.3	89.3	30.5	34.1	150000
12/29/1999	03:15	8.3	4.19	12.8	37.6	89.7	30.6	34.1	195000
12/27/1999	02:00	15	2.61	8.4	24.2	92.9	32.3	34.7	117000
12/25/1999	04:00	12.3	2.97	9.5	27.4	92	31.8	34.6	110000
12/24/1999	08:20	8.6	2.85	9.2	26.6	93.1	32.4	34.8	80000
12/23/1999	15:30	9.1	3.06	9.9	28.4	92.9	32.2	34.6	75000
12/21/1999	04:00	9	3.56	11.5	33.5	94.1	32.5	34.5	90000
12/19/1999	05:00	8.3	3.59	11.7	33.6	93.7	32.7	34.9	75000
12/18/1999	06:00	10	3.33	10.9	31.1	93.4	32.7	35	75000
12/17/1999	03:00	9.1	2.91	9.3	27.3	93.7	32	34.1	45000
12/13/1999	04:00	11.1	3.34	10.8	31.5	94.2	32.4	34.4	60000

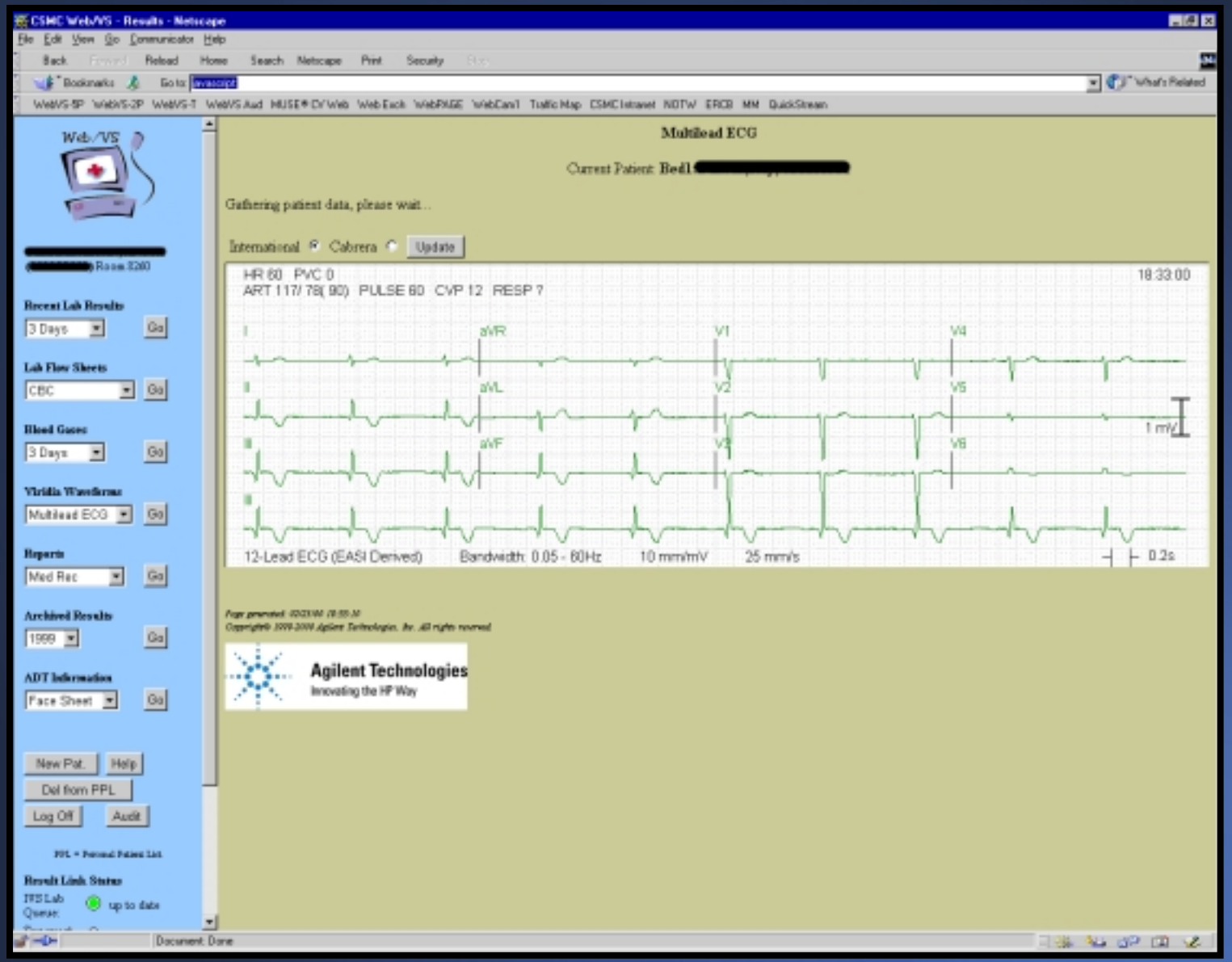
Web/VS: Transcribed Reports



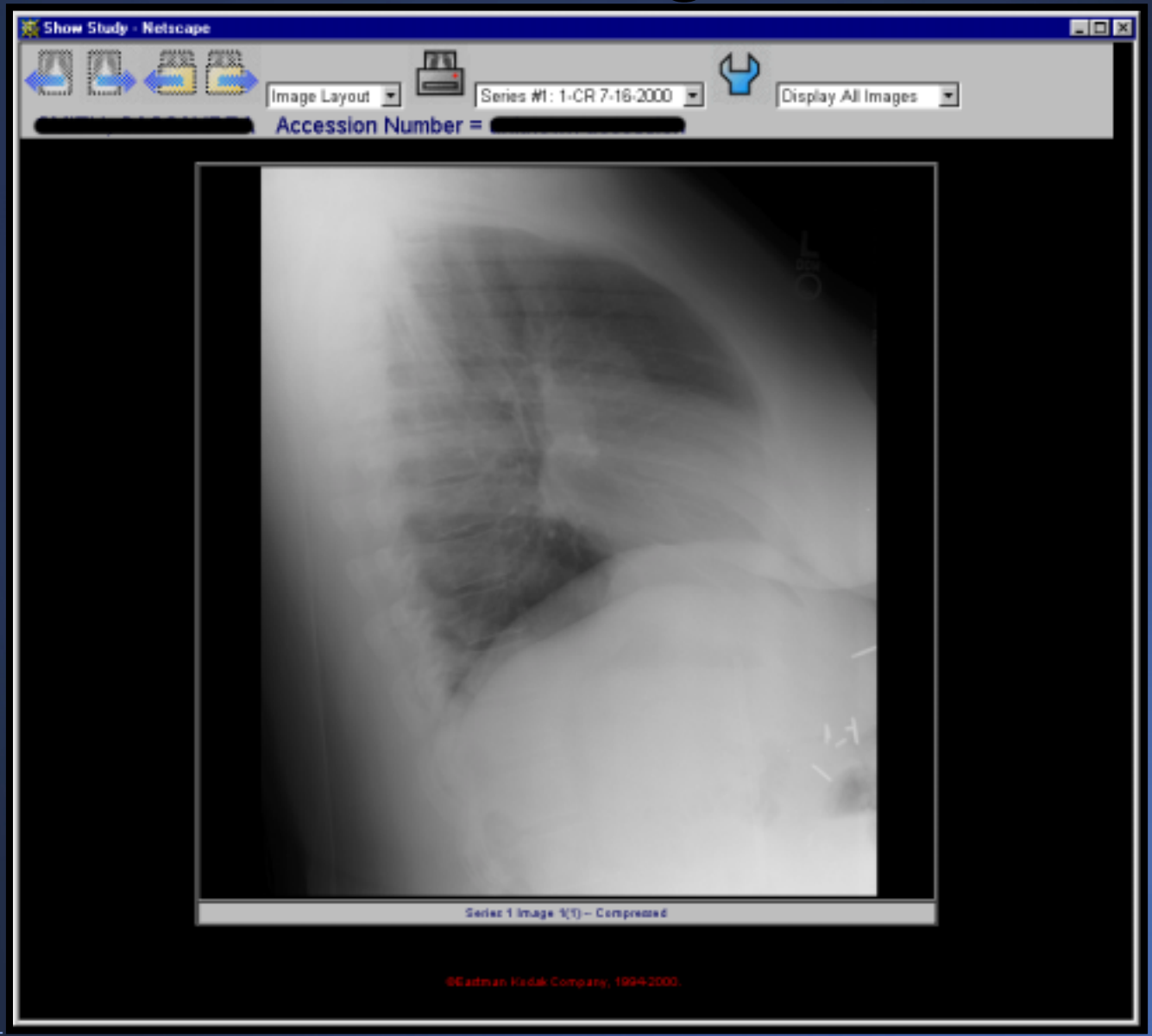
Web/VS: ECG Reports



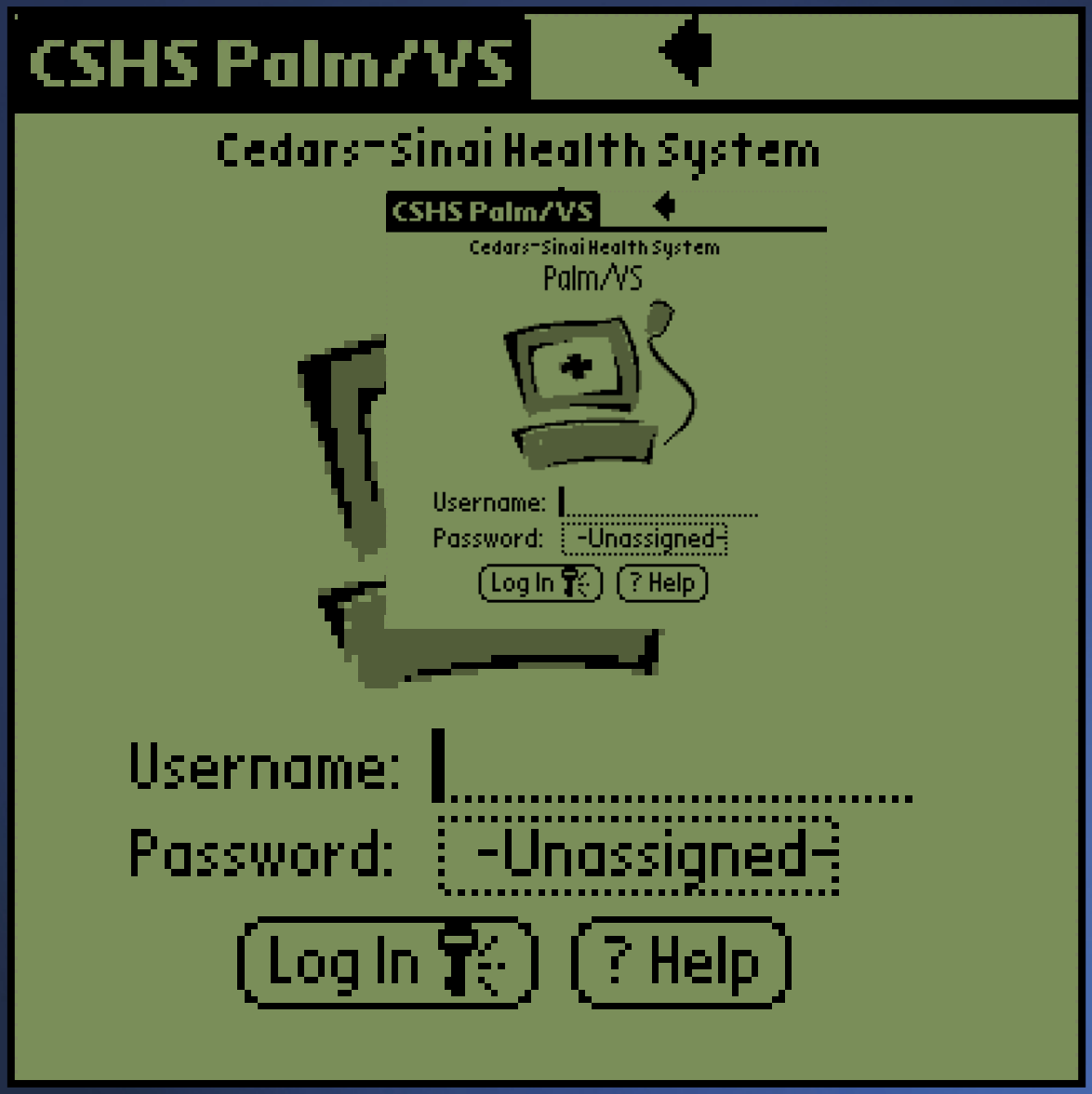
Web/VS: ICU Monitor Waveforms



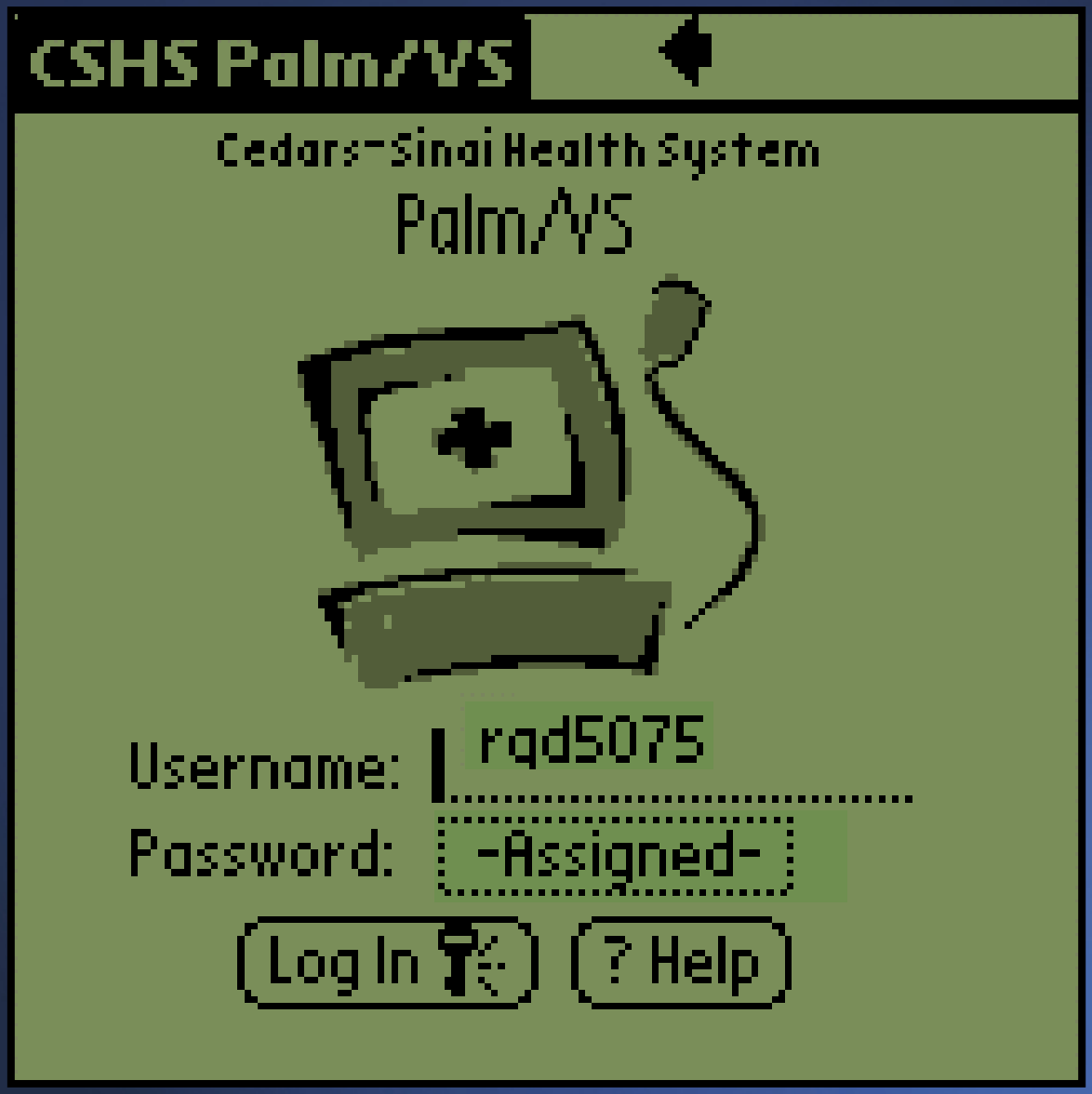
Web/VS: PACS Images



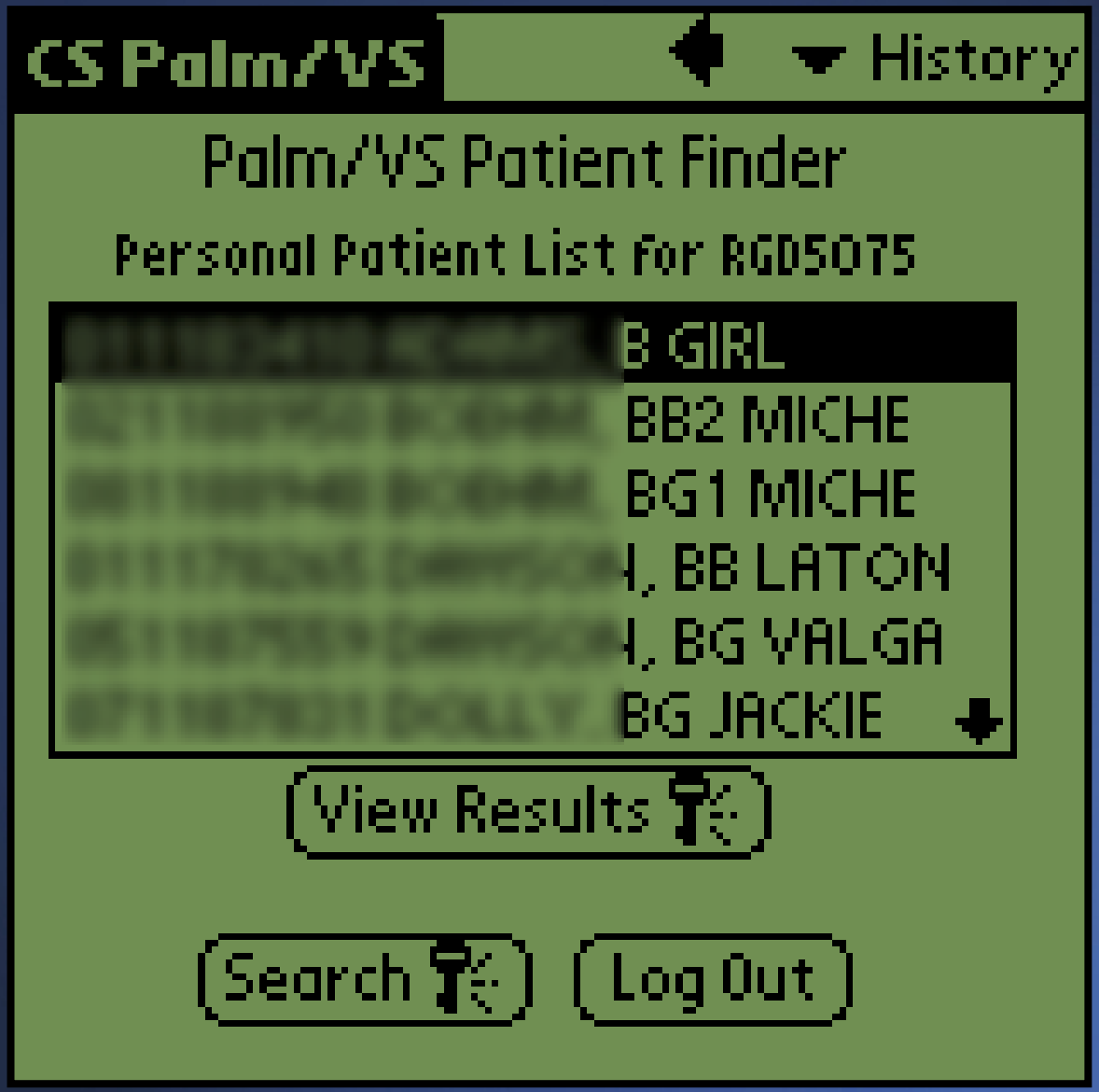
Palm/VS: Clinical Data Viewer



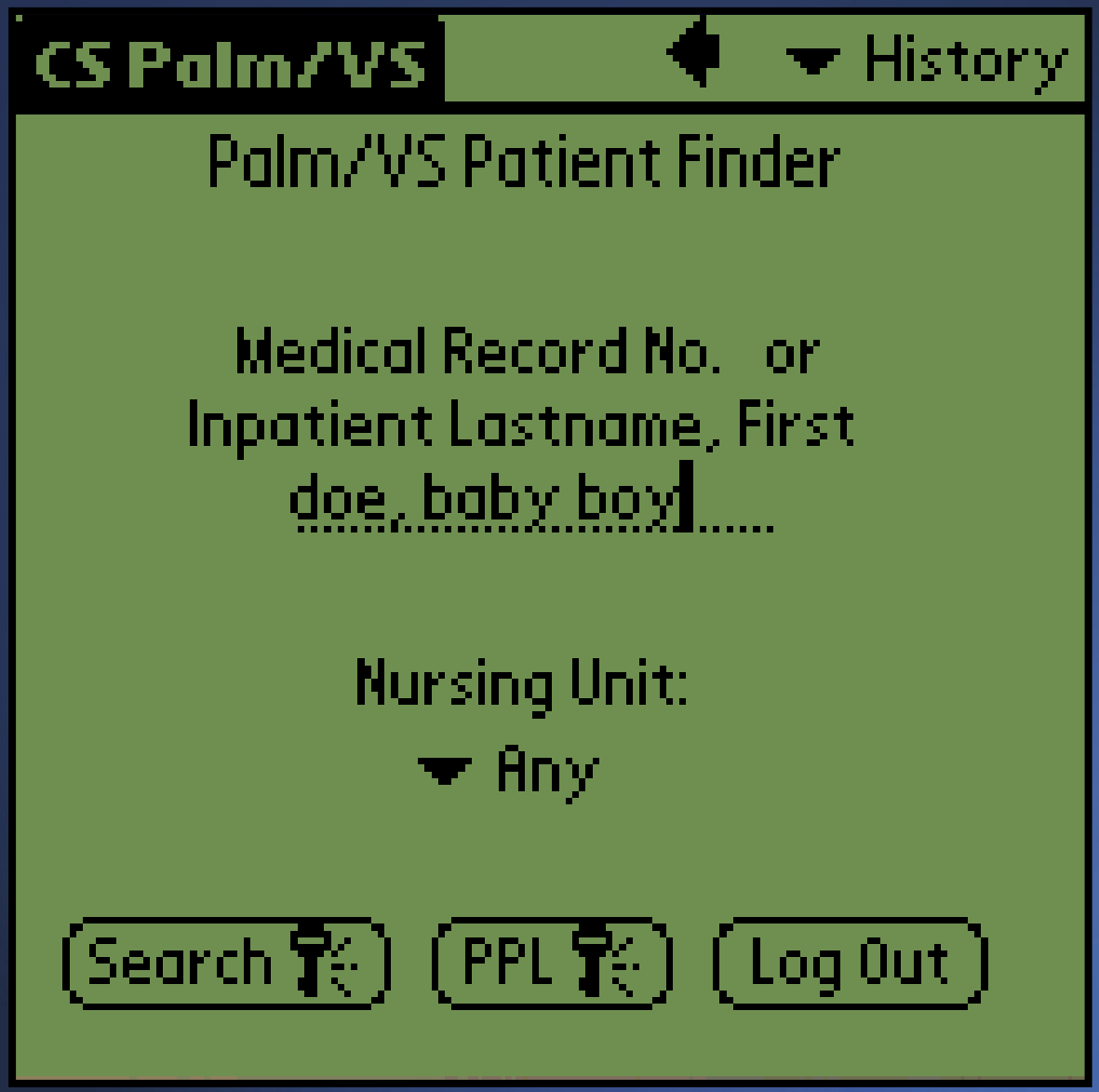
Palm/VS: Log In



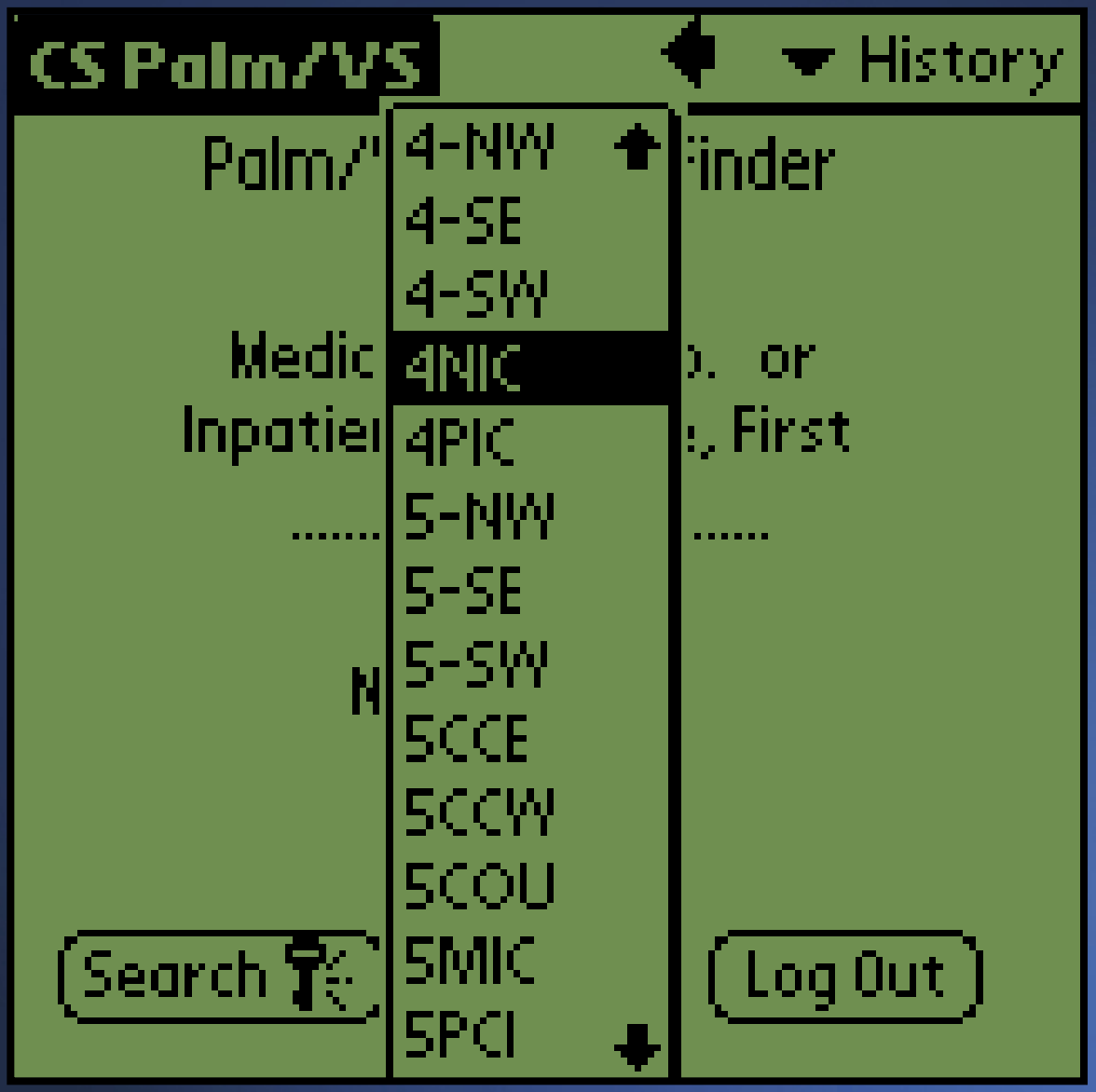
Palm/VS: Personal Patient List



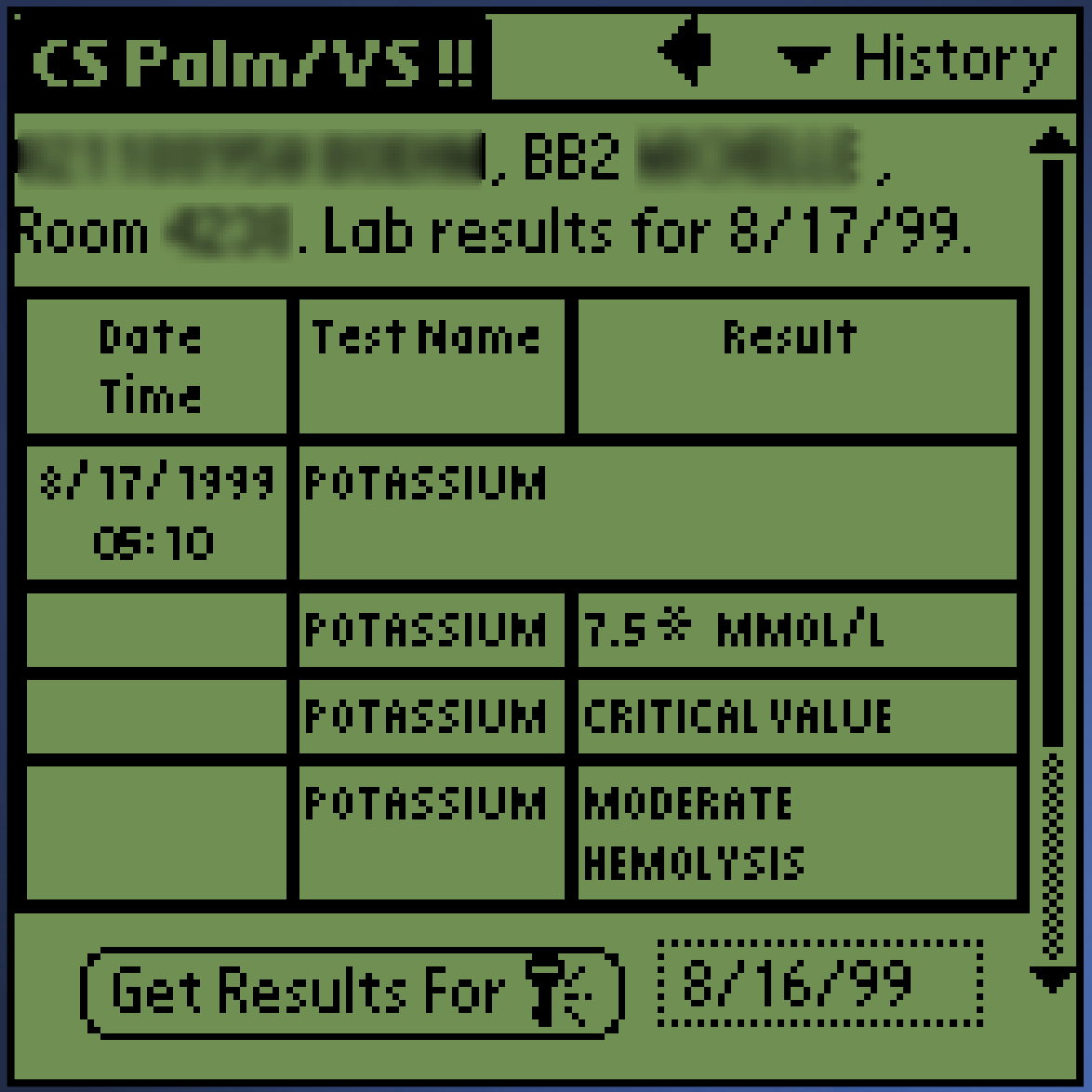
Palm/VS: Search by Name



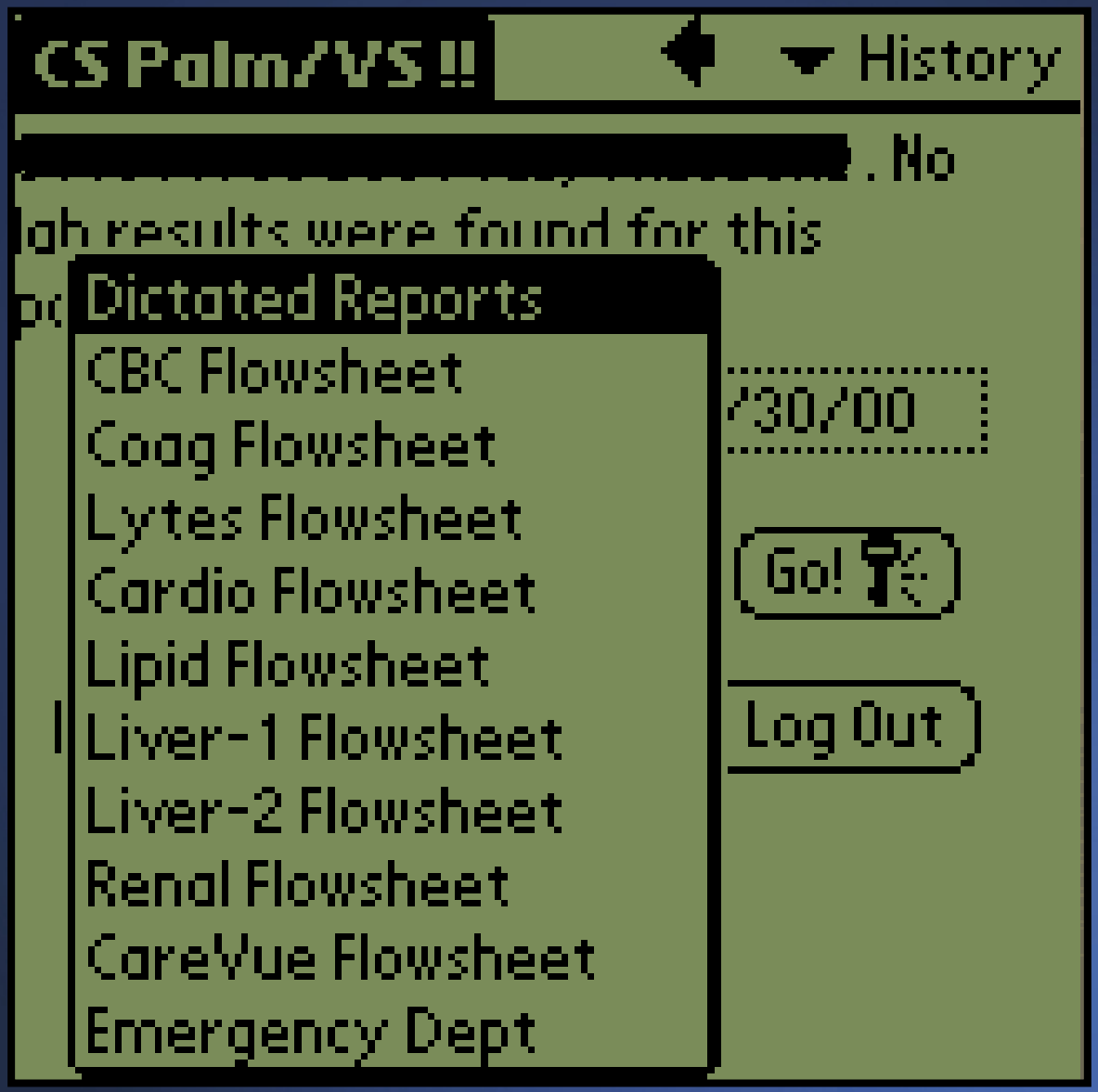
Palm/VS: Search by Unit



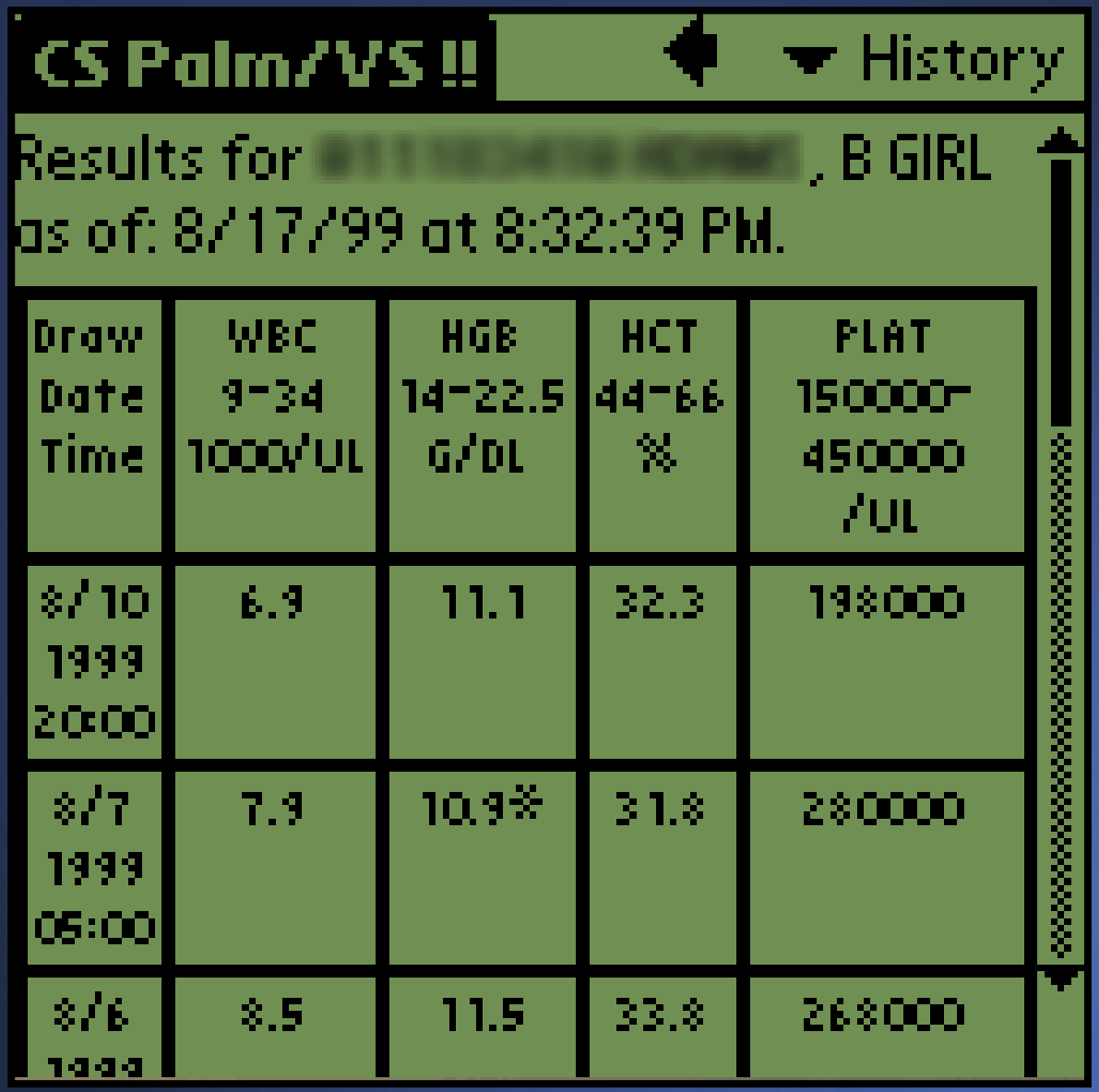
Palm/VS: Recent Results



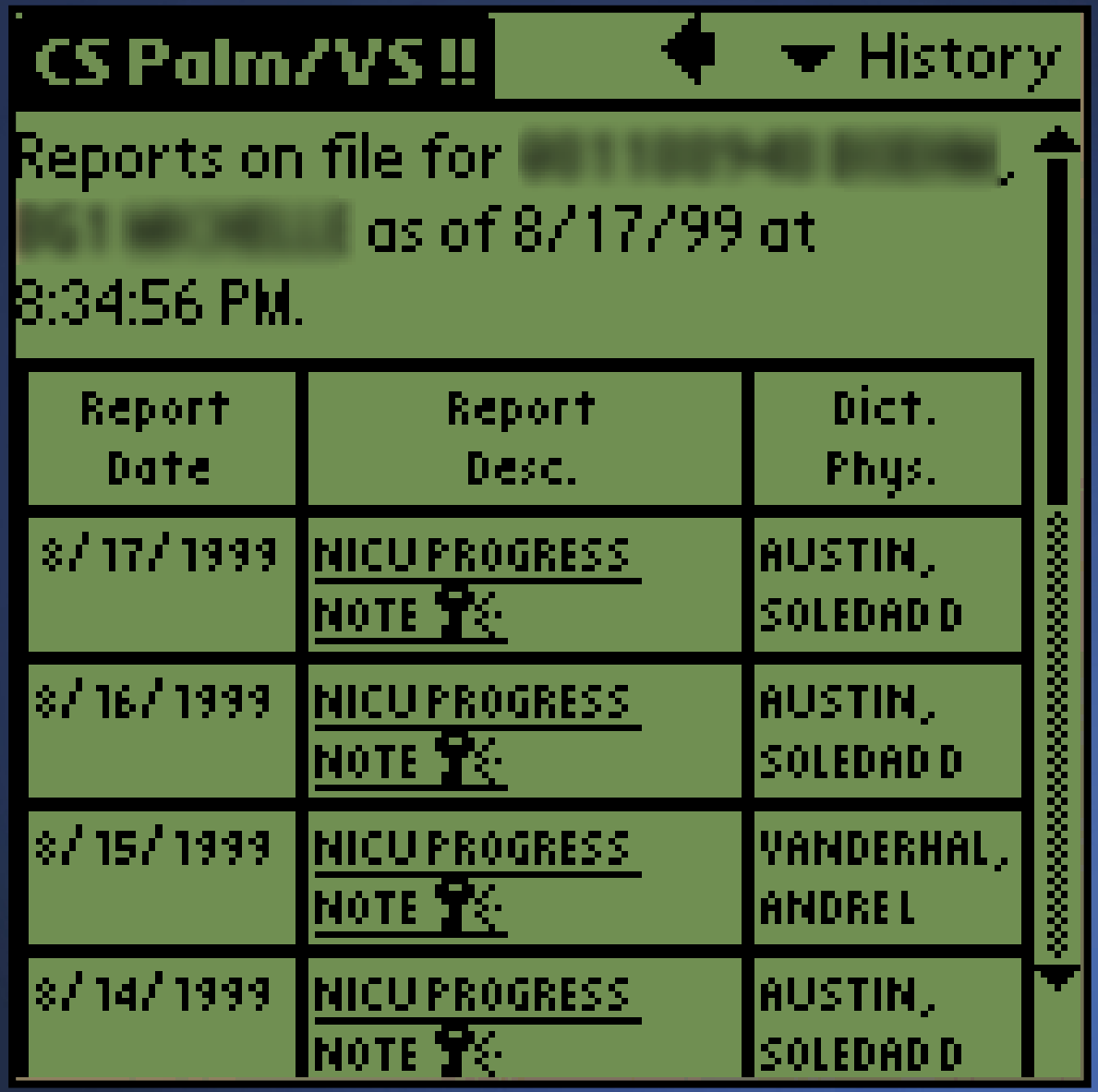
Palm/VS: Reports Menu



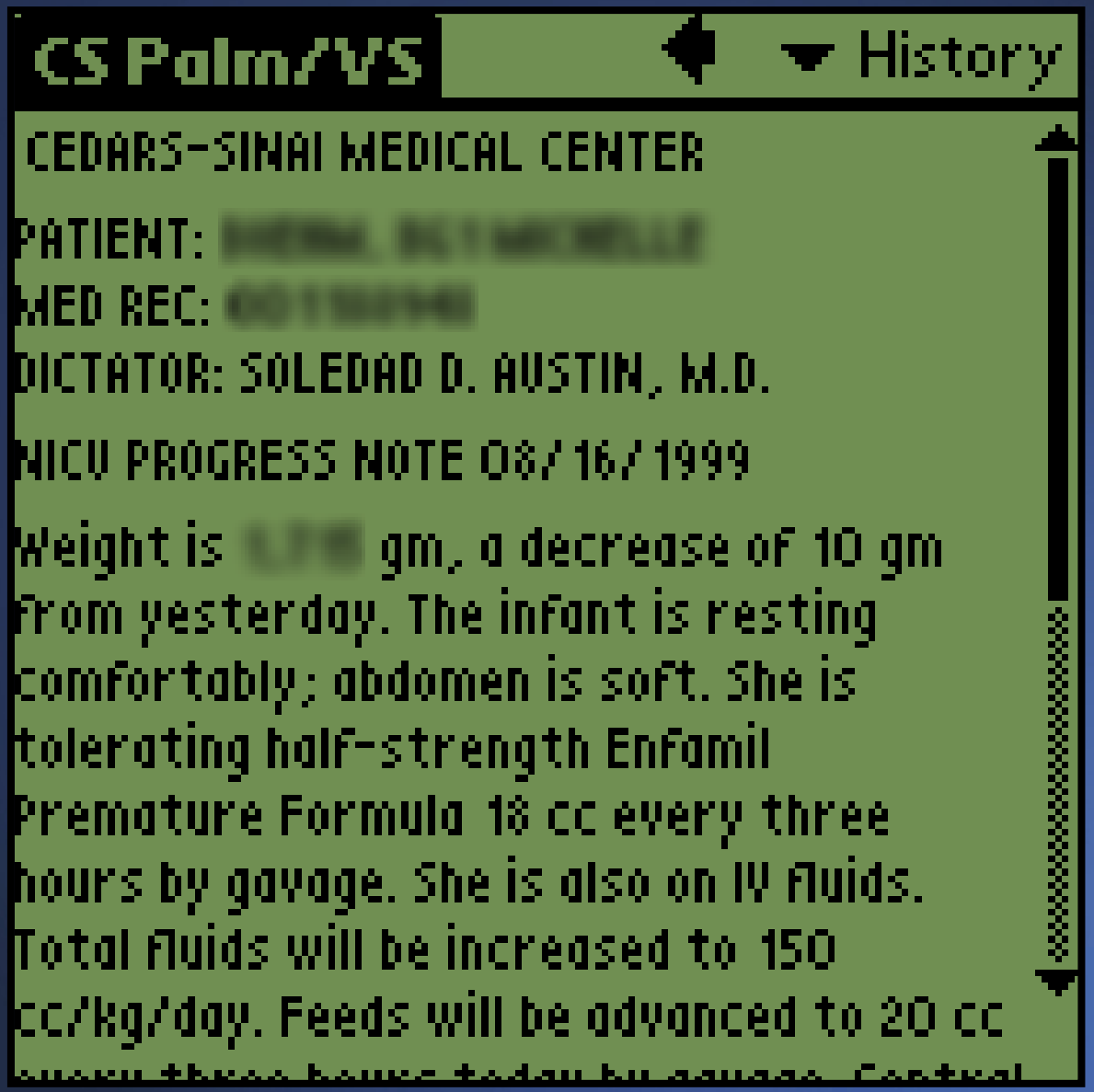
Palm/VS: Sample Flowsheet



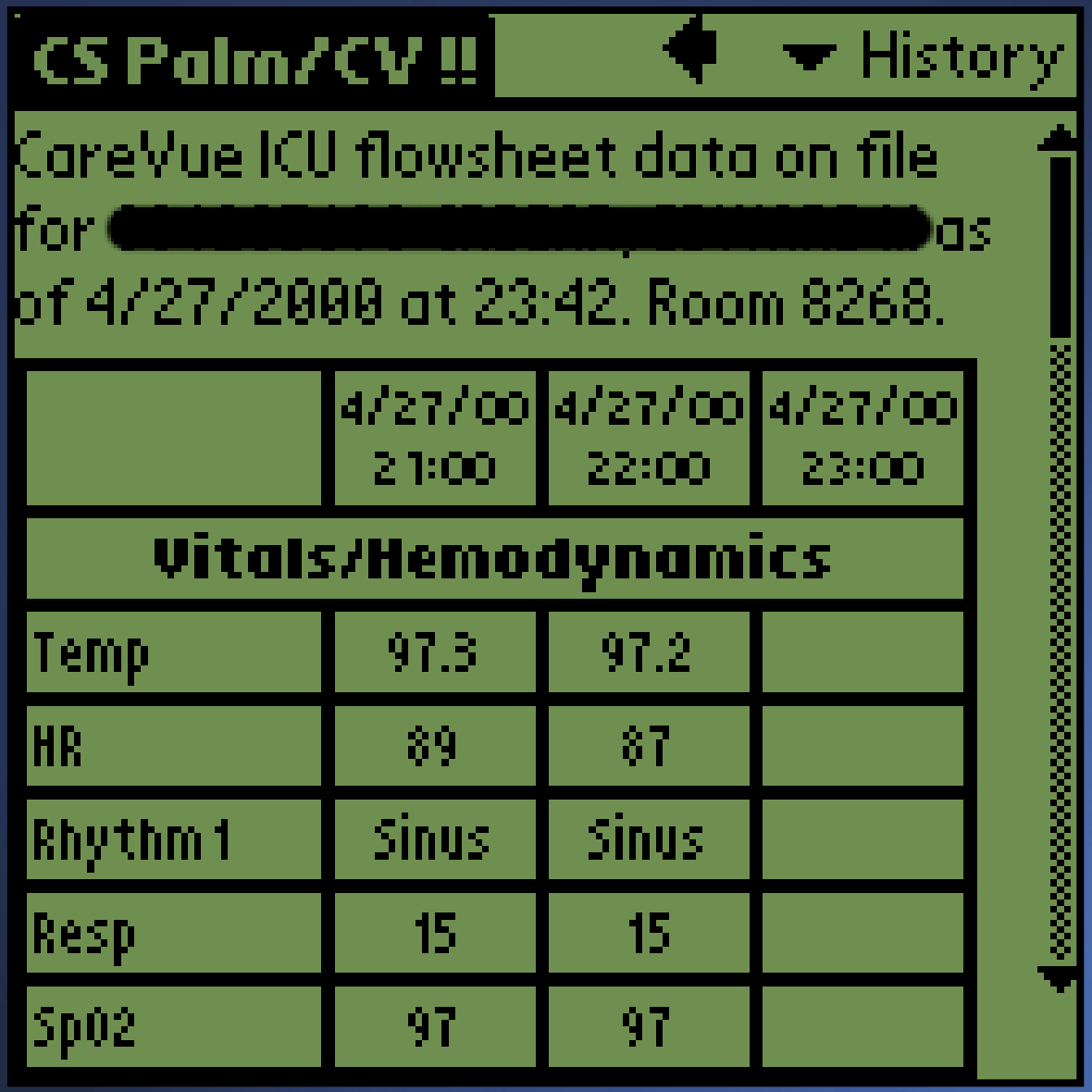
Palm/VS: Transcribed Report List



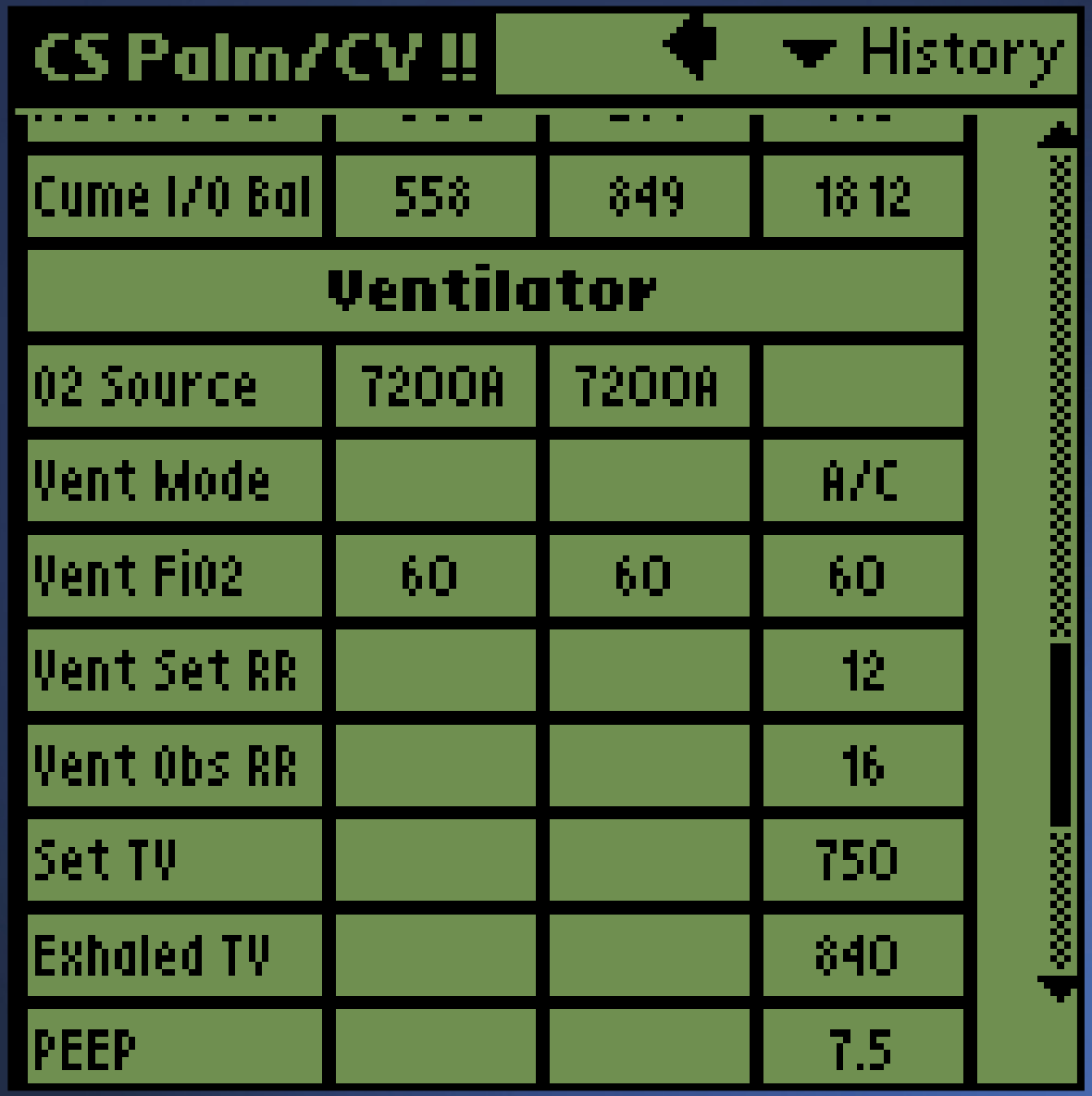
Palm/VS: Transcribed Report Detail



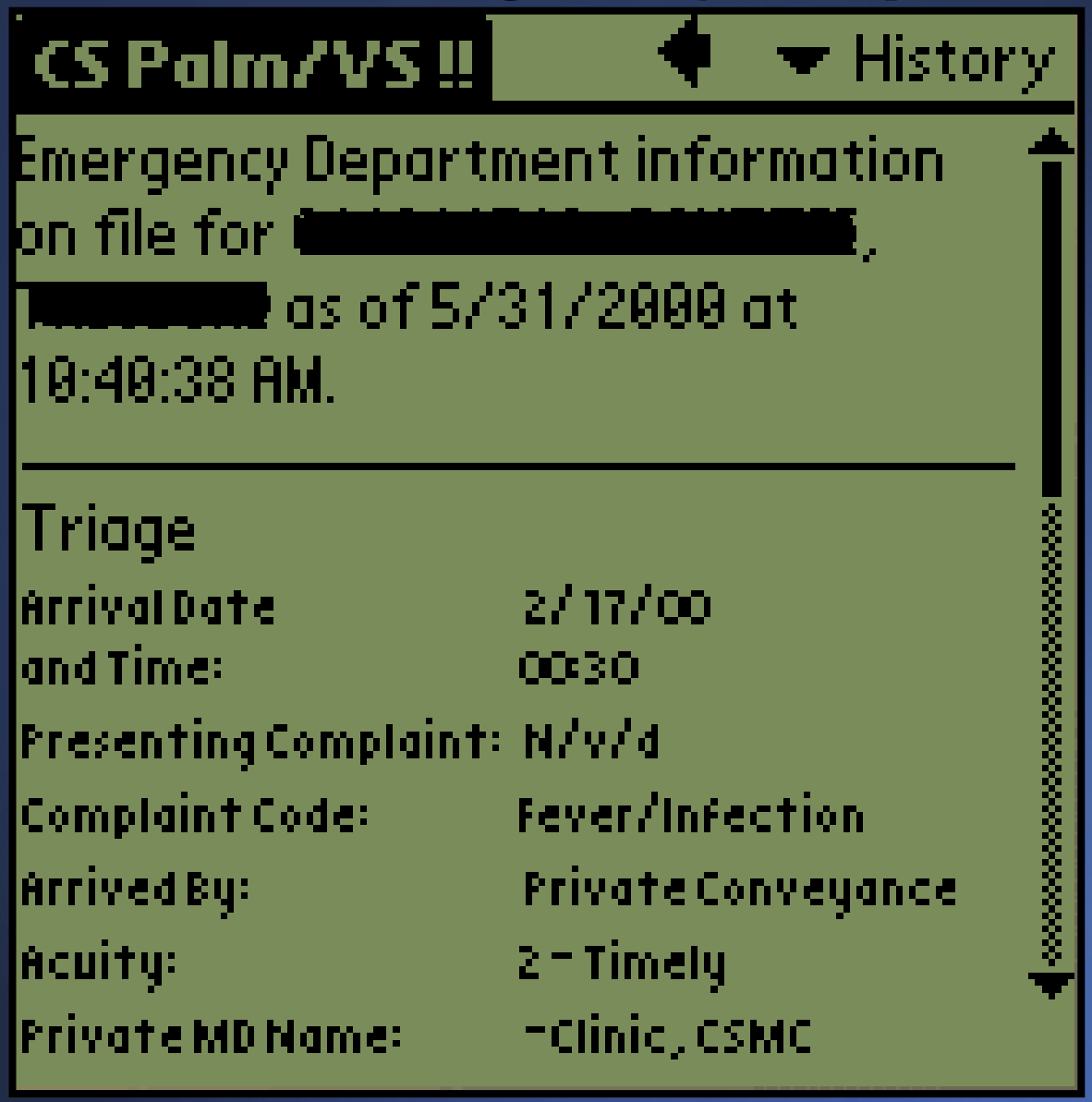
Palm/VS: ICU Vital Signs



Palm/VS: ICU Ventilator Settings



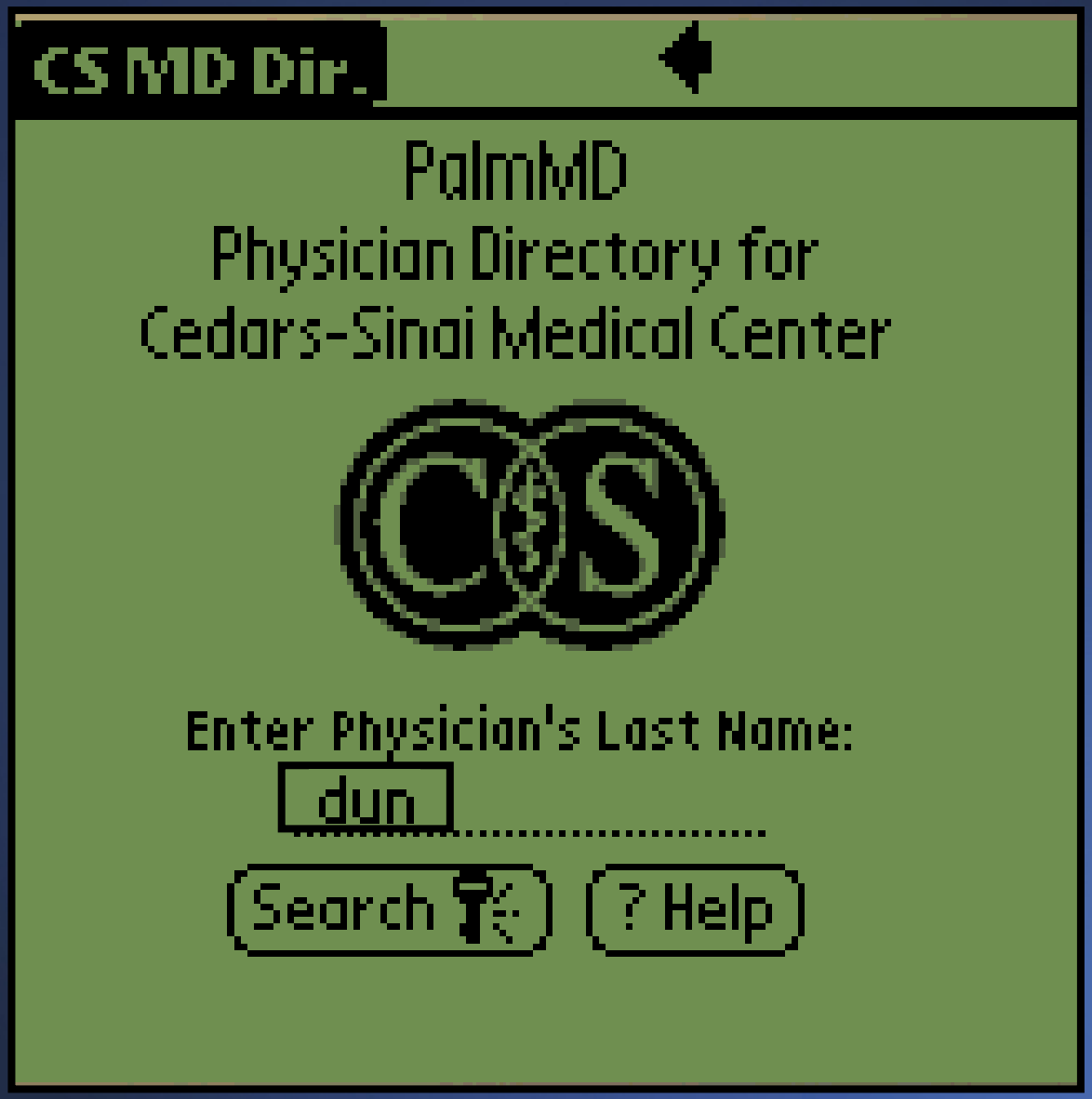
Palm/VS: Emergency Dept.



Other Wireless Applications Provided to CSMC Physician Users

- Developed in-house:
  - Attending Staff Directory
  - Employee Directory
  - Library Document/Book Orders
  - Alphanumeric Paging
- Third-party:
  - MedLine
  - Go Translator
  - POP3 Email Client

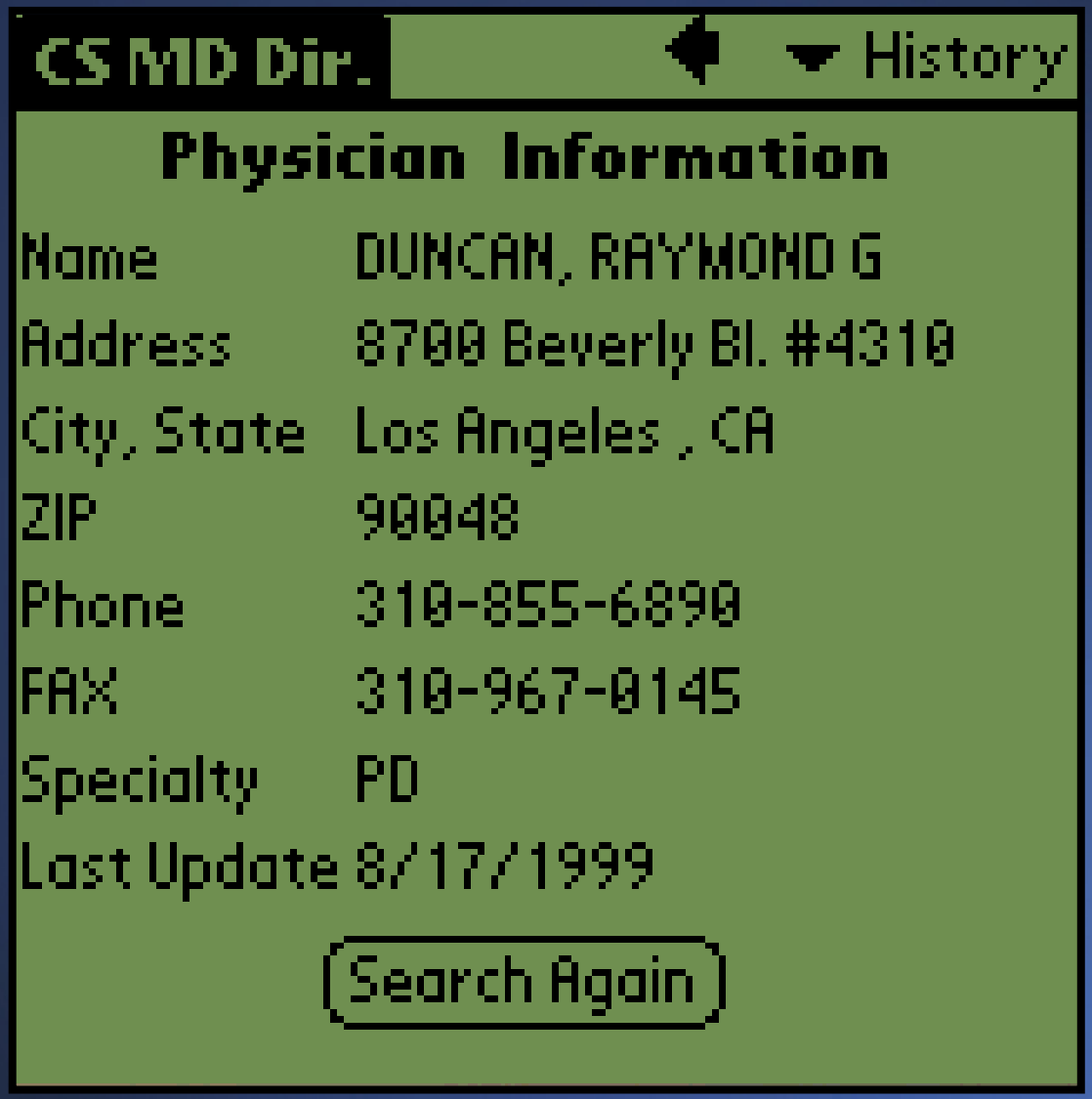
Palm/MD: Physician Directory



Palm/MD: Name Search Results



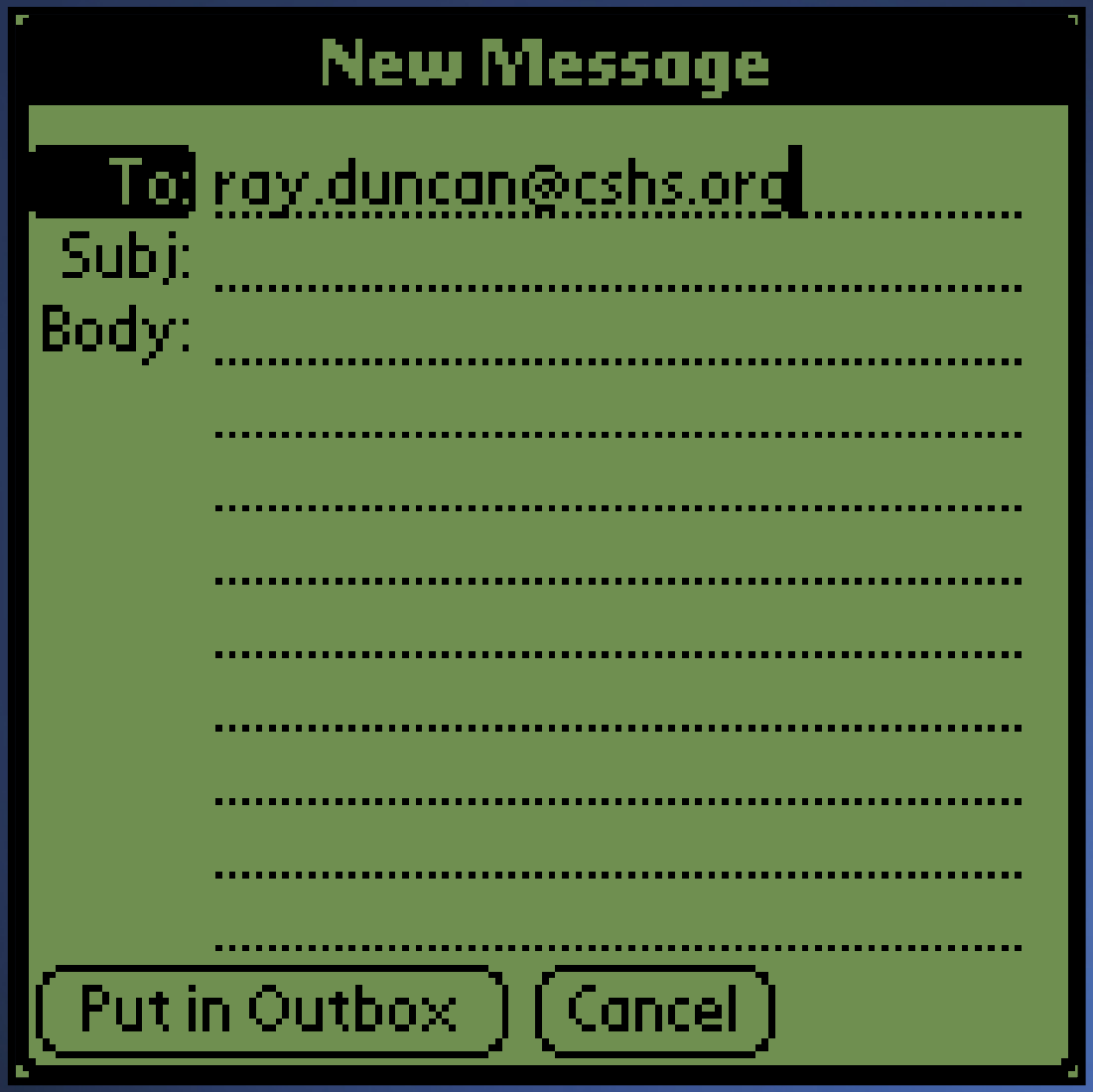
Palm/MD: Detailed Info



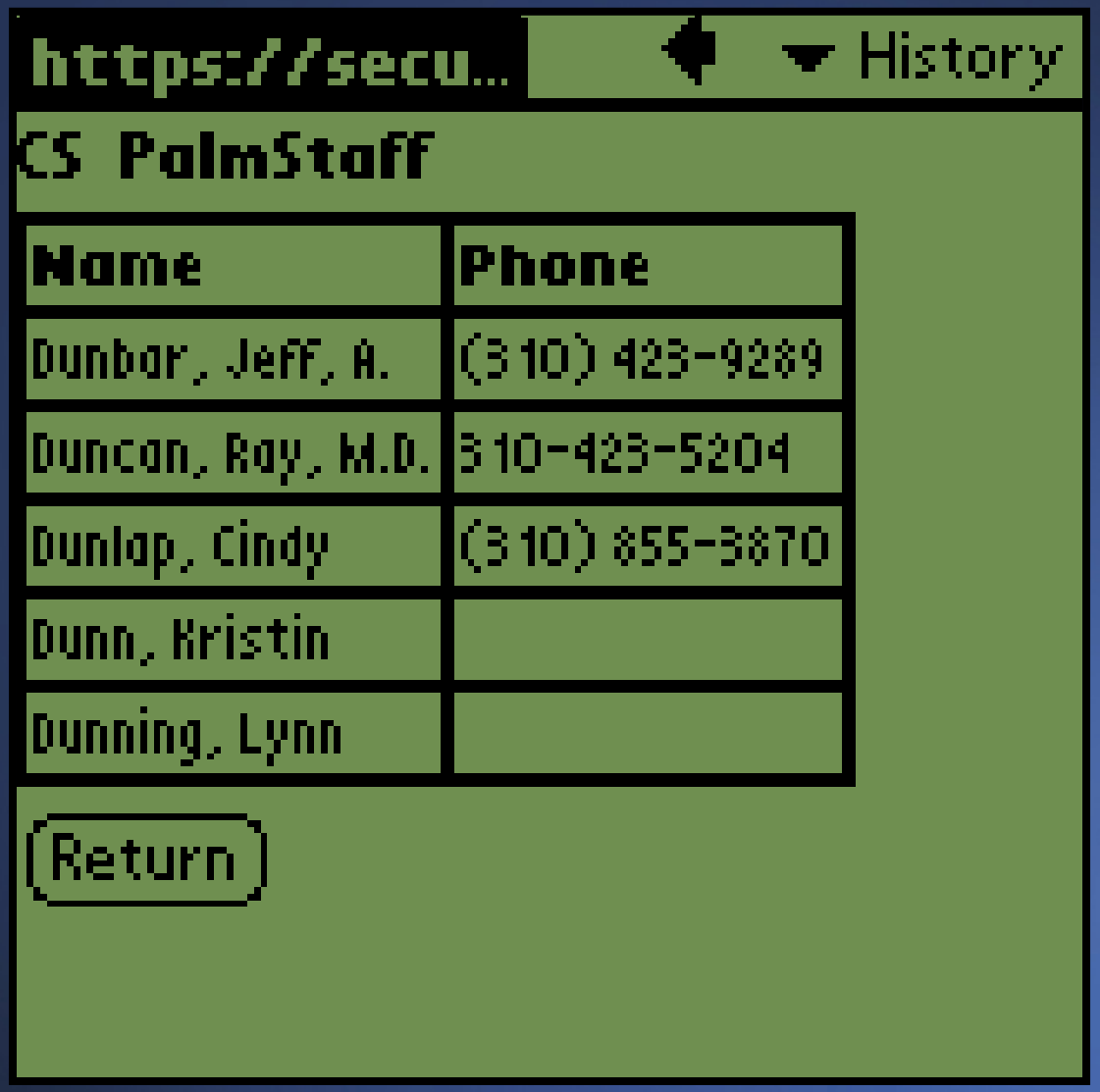
Palm/Staff: Employee Directory



Palm/Staff: Email Lookup



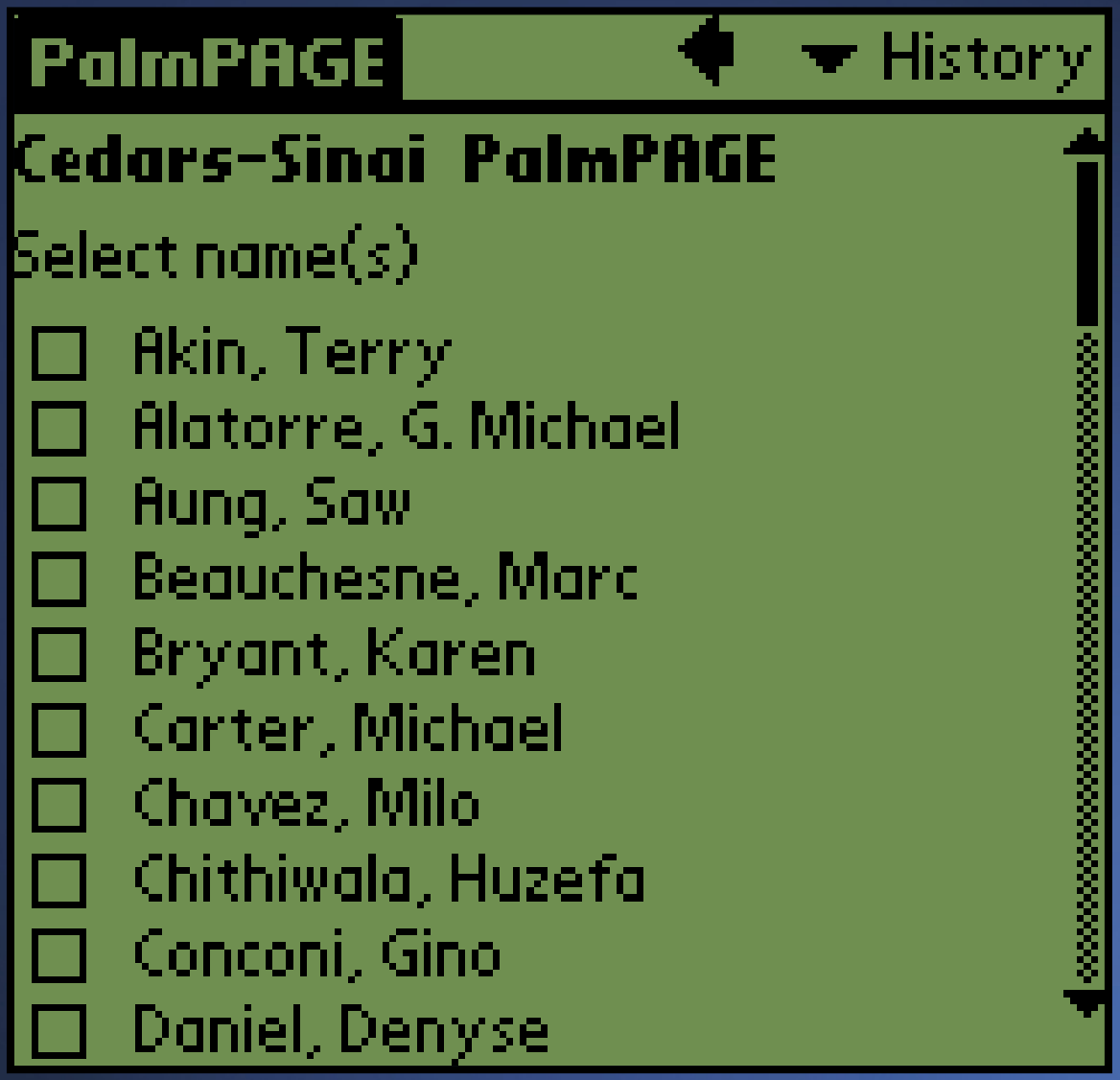
Palm/Staff: Phone Lookup



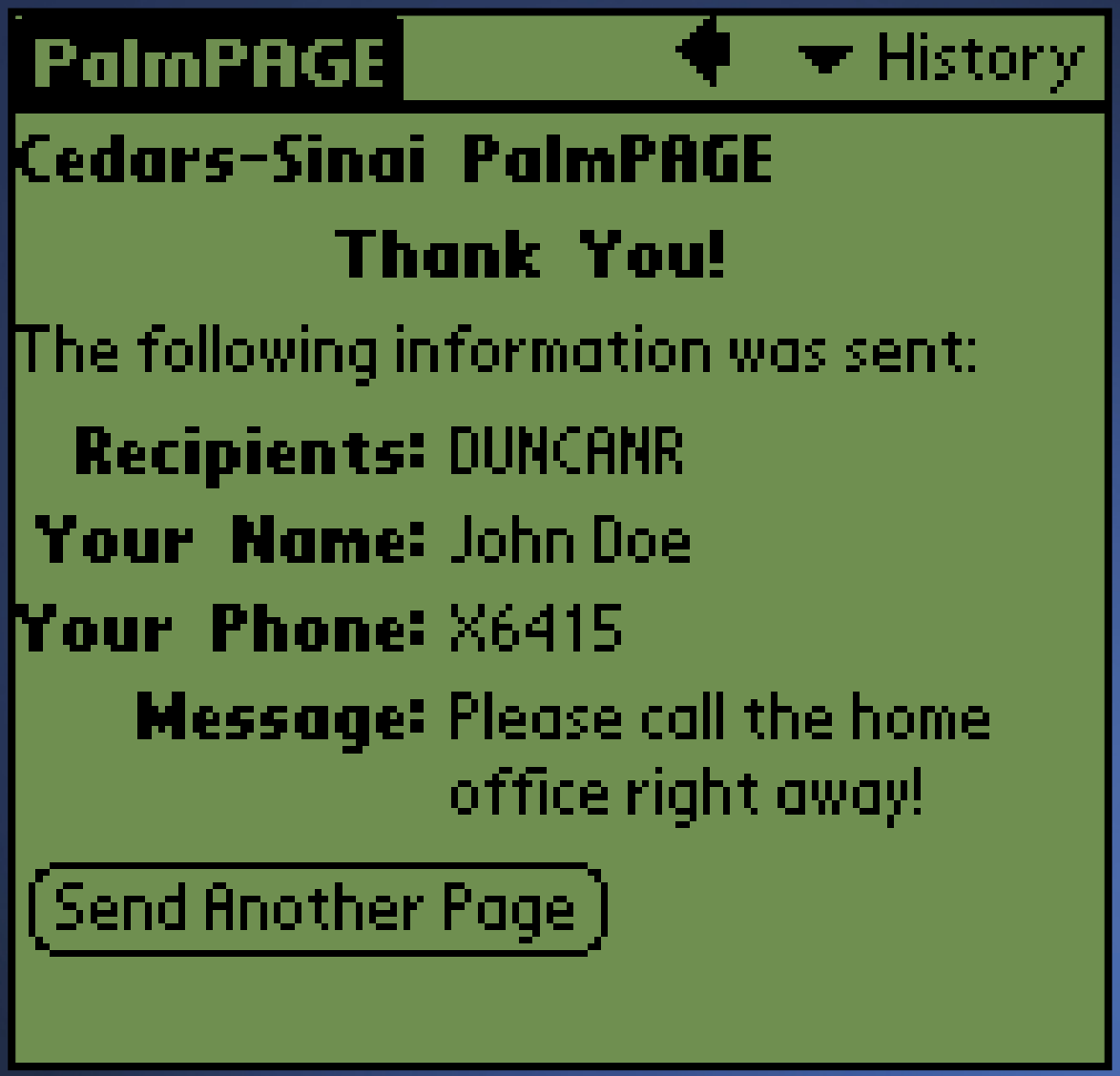
Palm/Page: Alphanumeric Paging



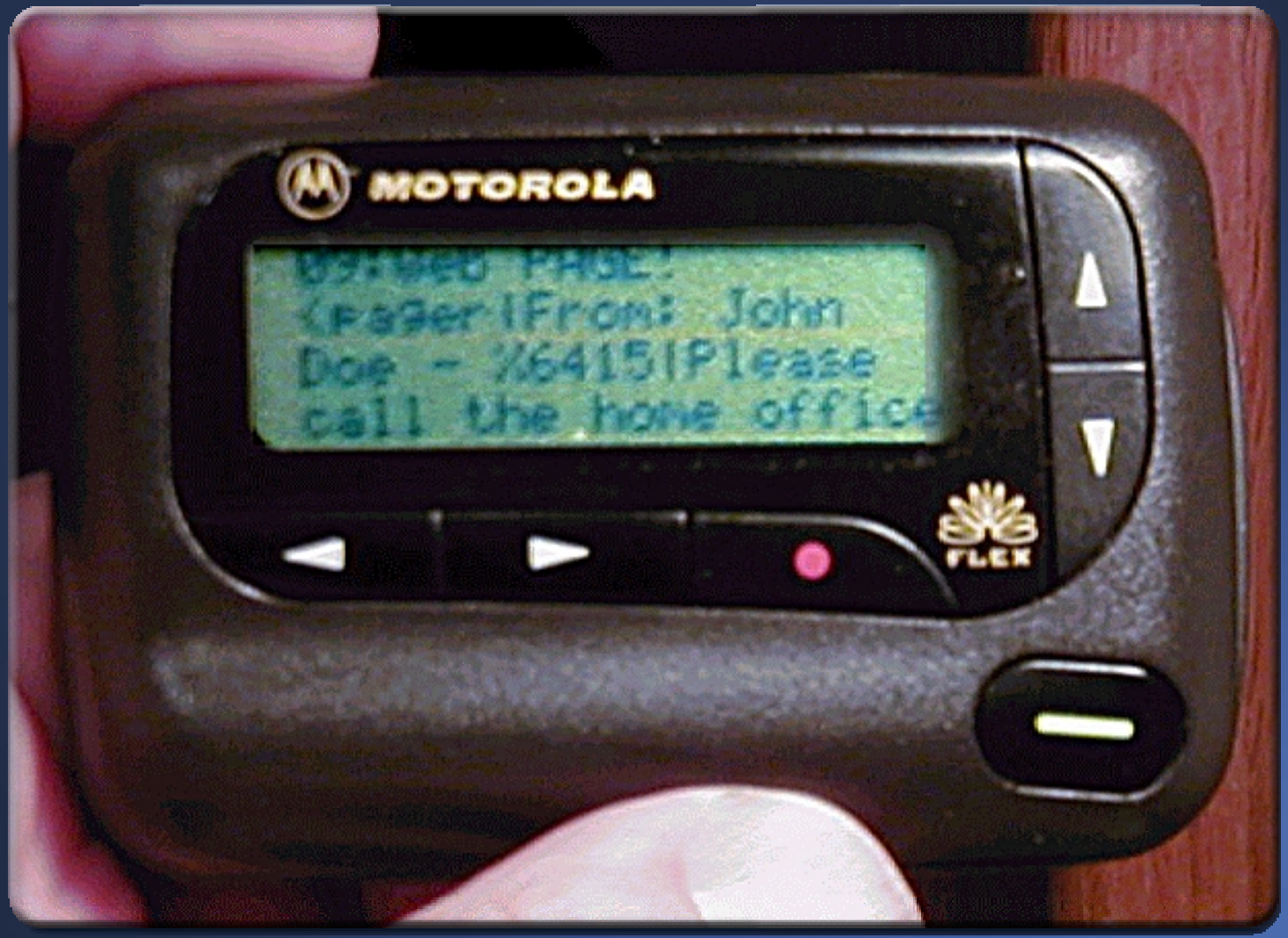
Palm/Page: On-Line Directory



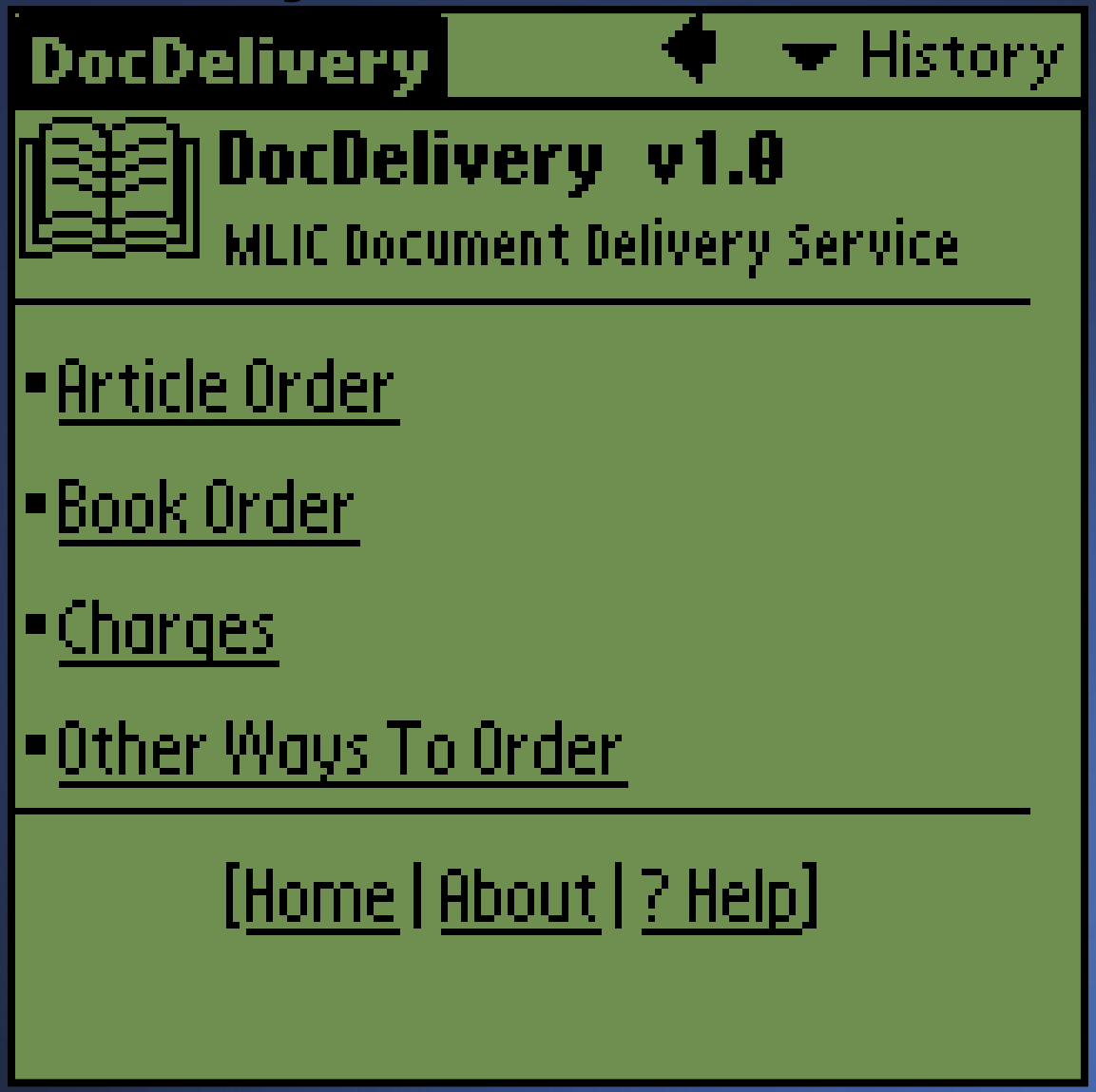
Palm/Page: Acknowledgement



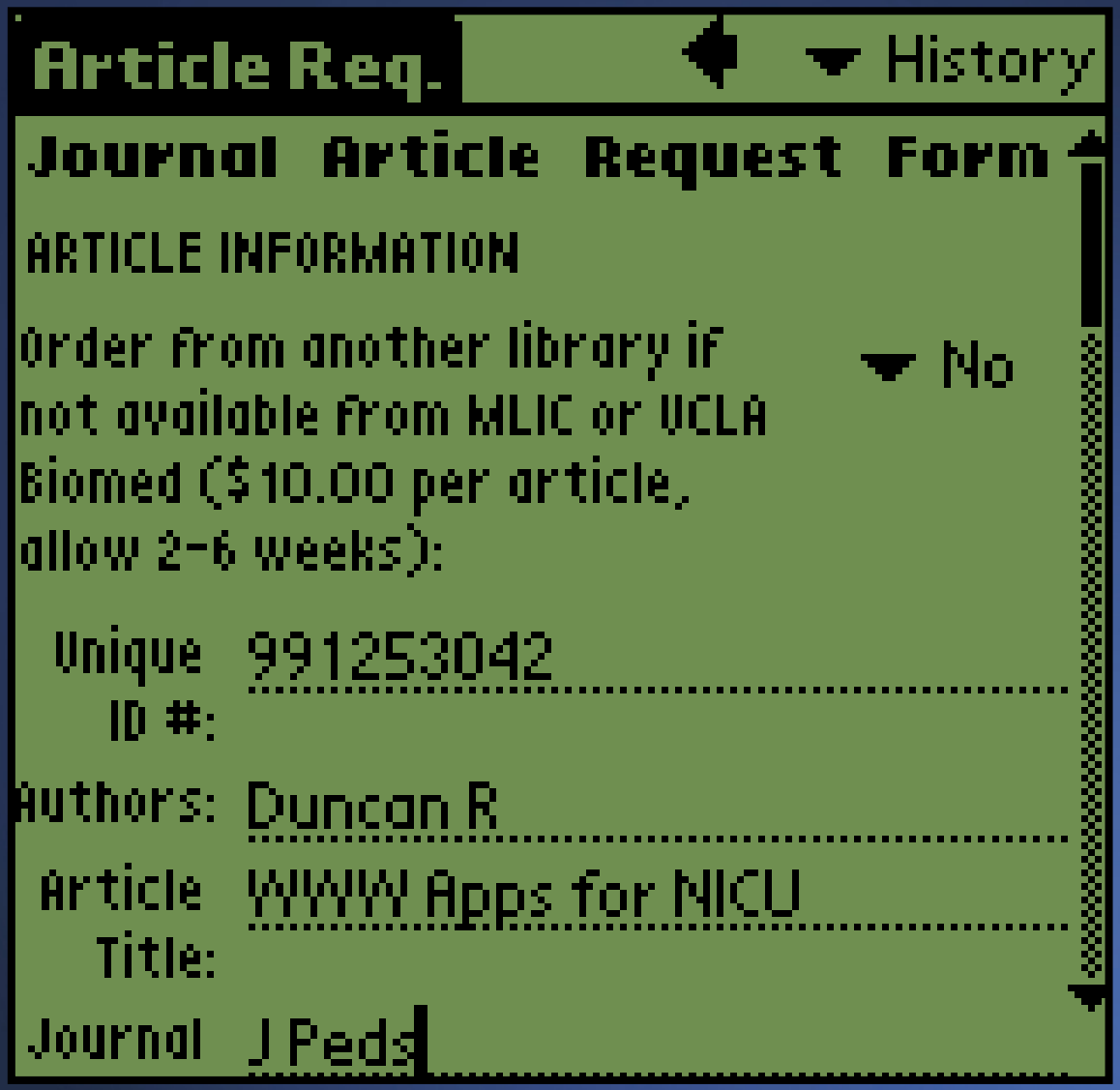
Palm/Page: Delivery



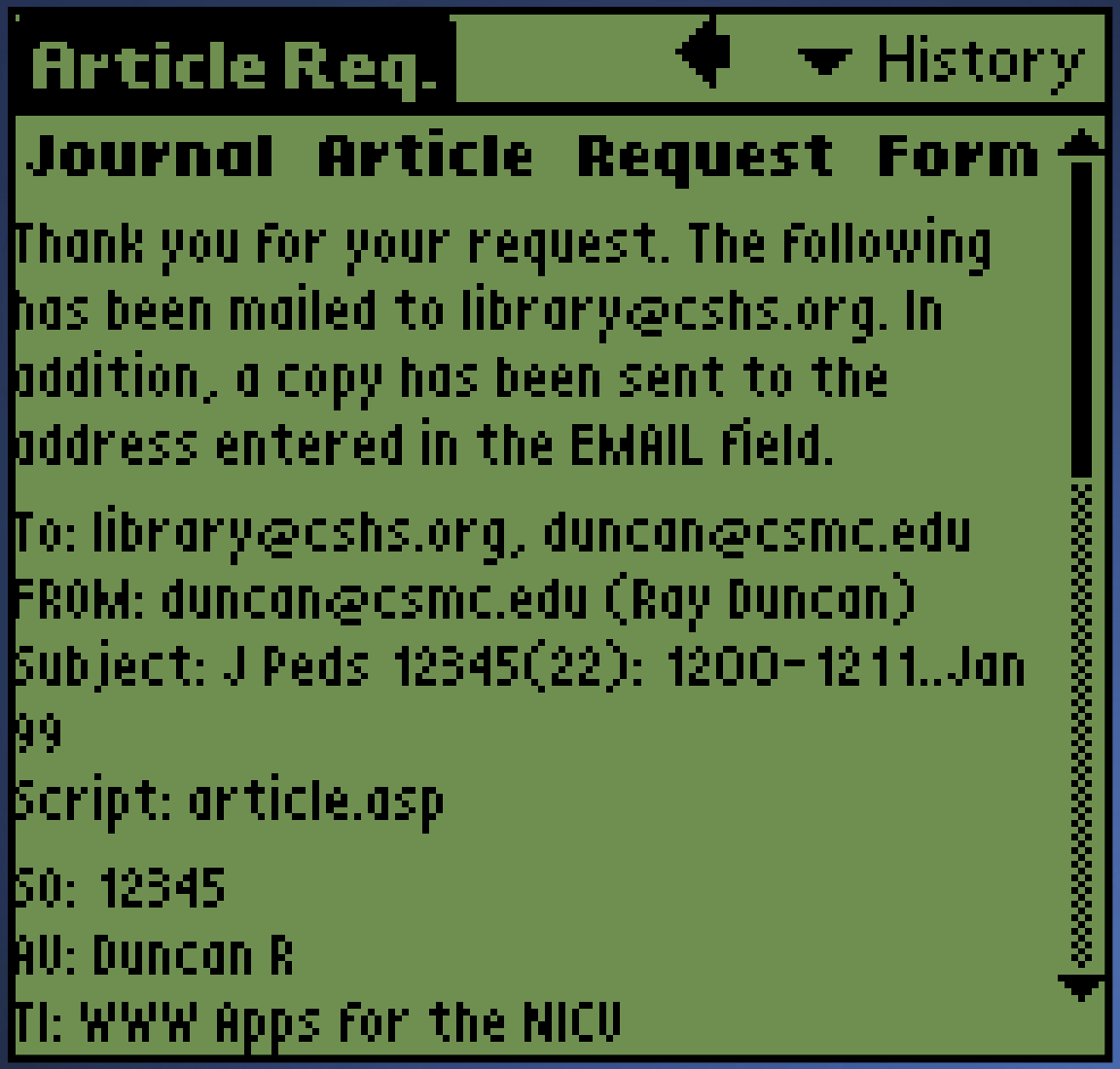
DocDelivery: Article and Book Orders



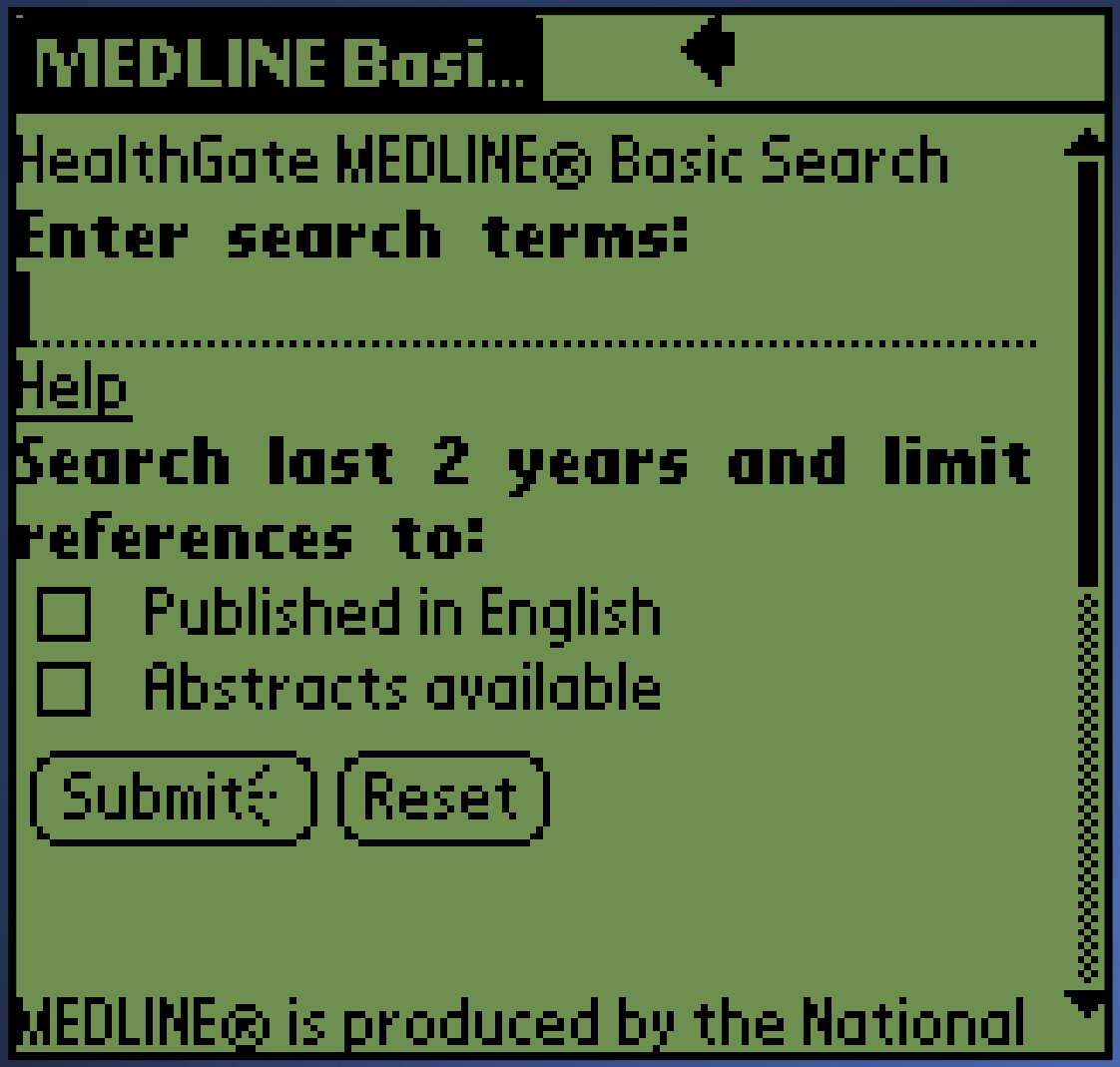
DocDelivery: Article and Book Orders



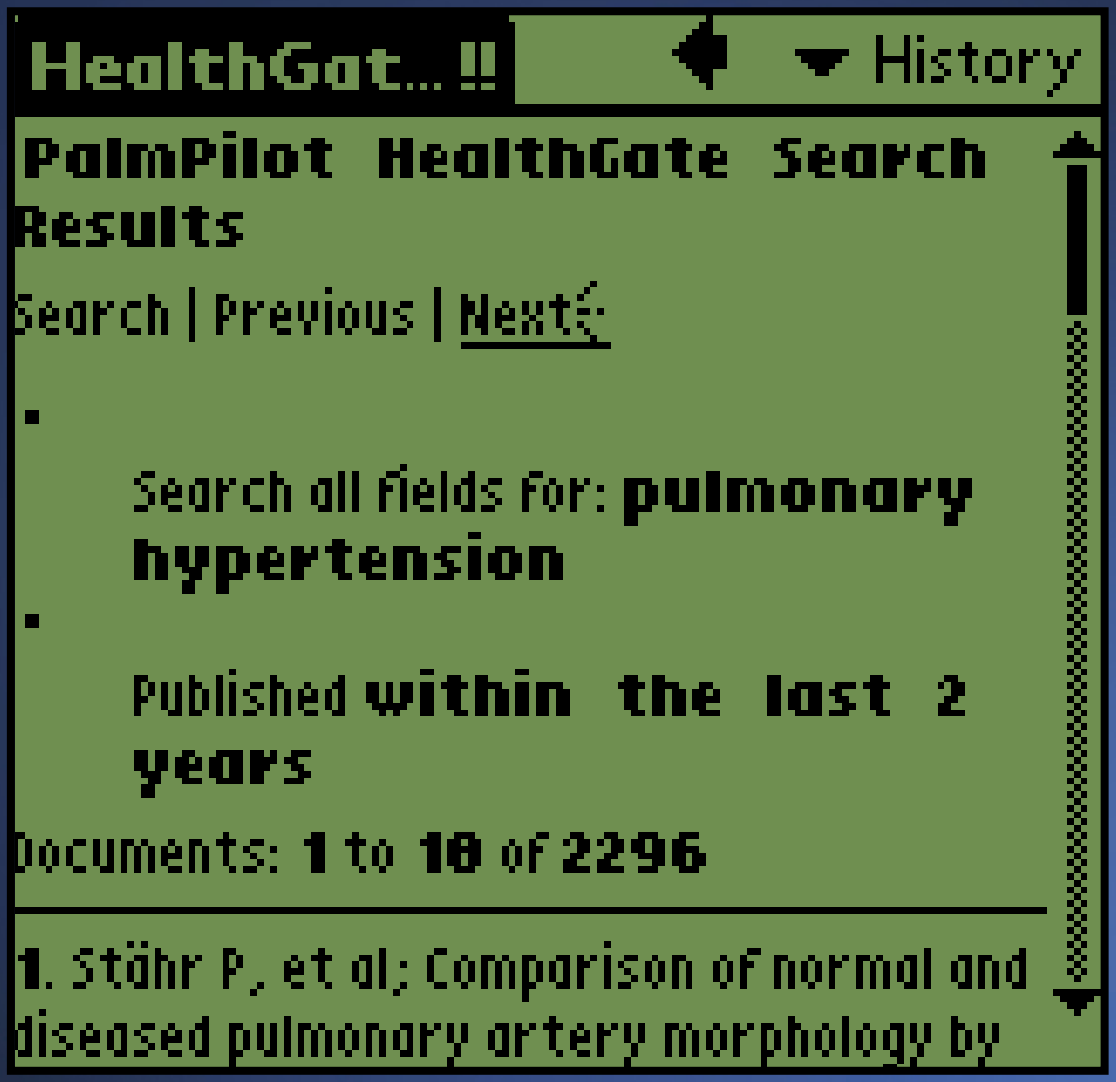
DocDelivery: Article and Book Orders



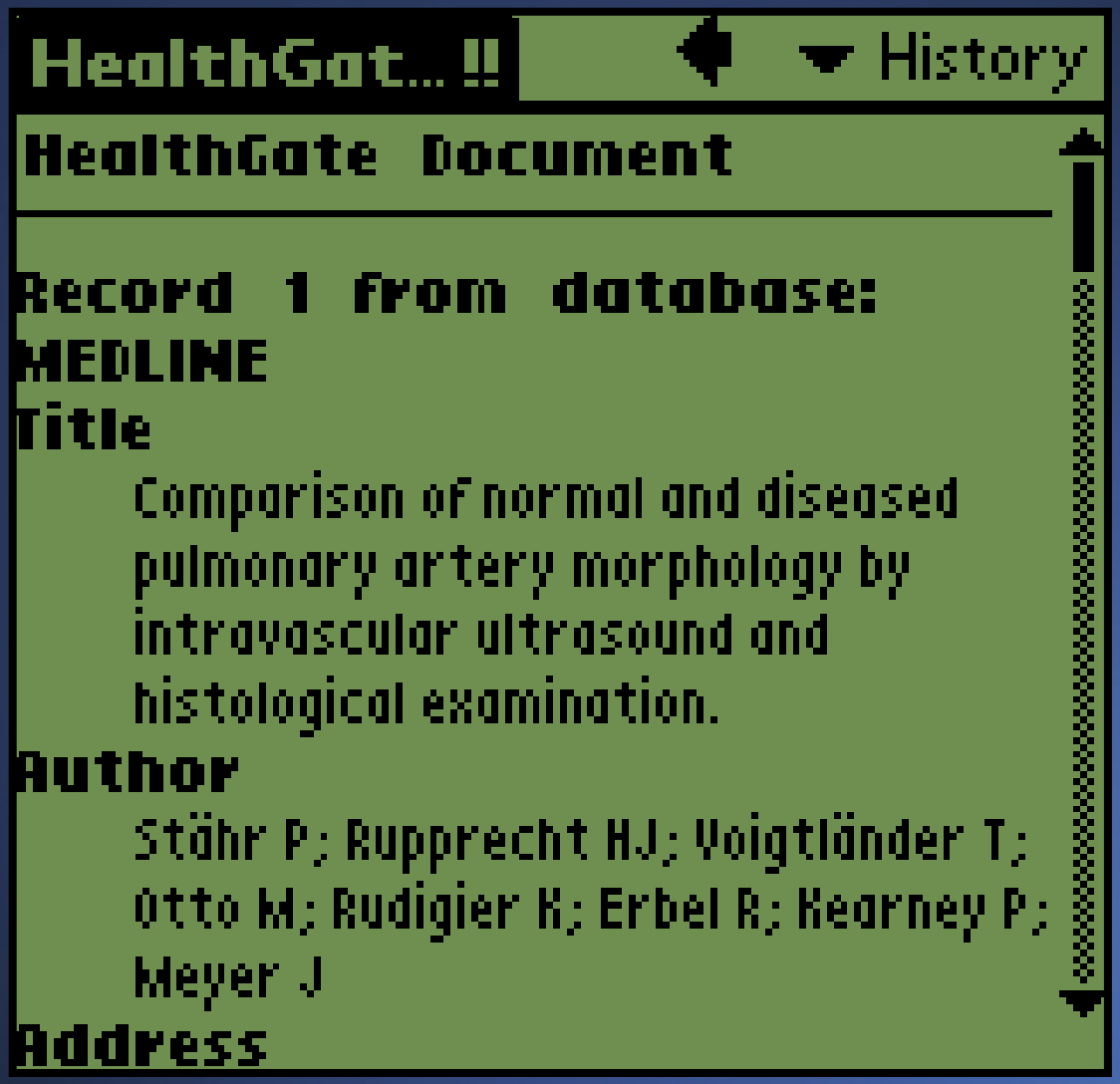
Free Wireless MedLine



Free Wireless MedLine



Free Wireless MedLine



Go! Translator



Go! Translator



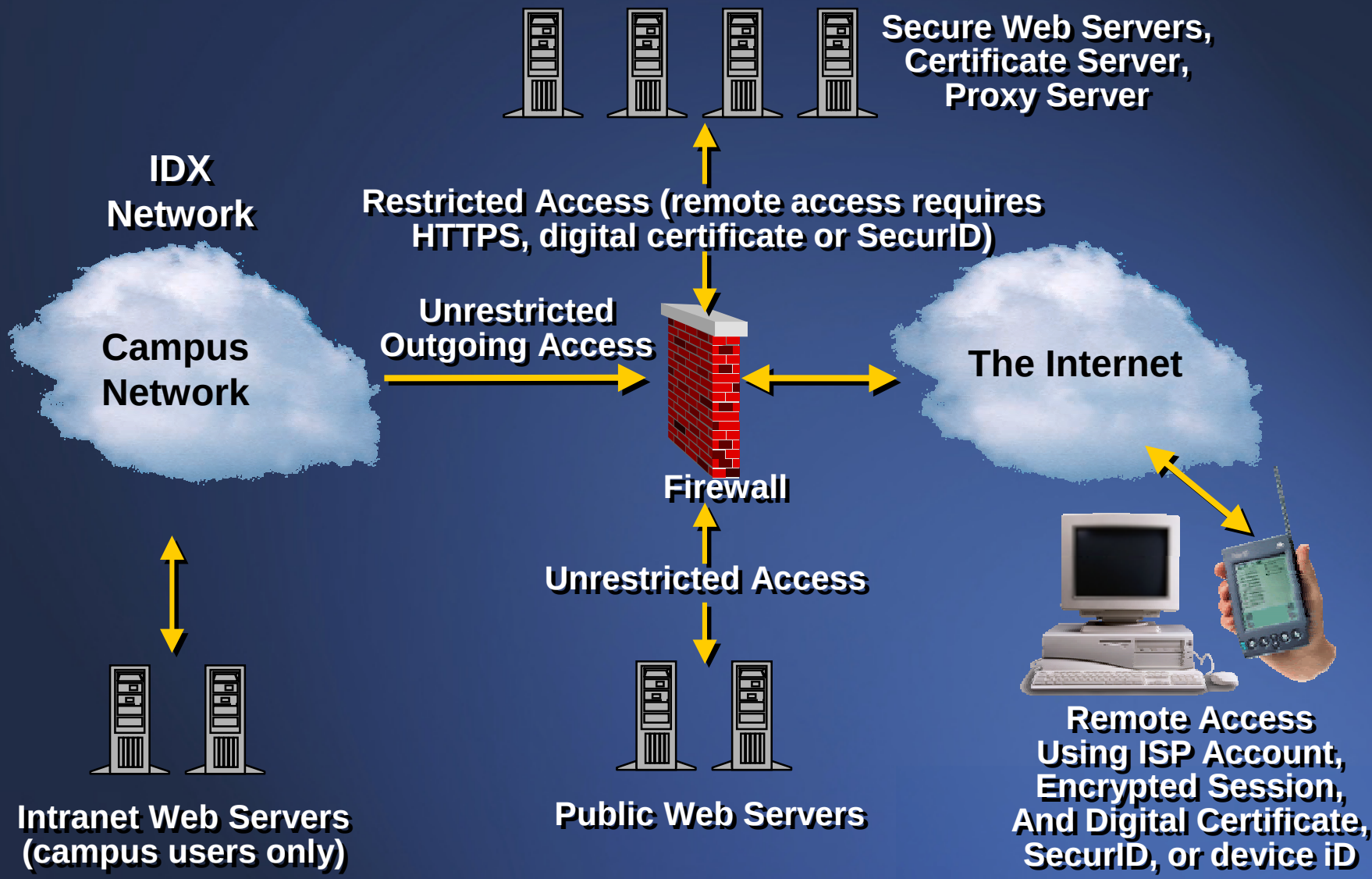
Security and Confidentiality

- Ease-of-Access Stance: Clinicians Can Access (nearly) All Data
- All Employees Sign the HR “Need to Know” Policy
- Close Coordination with Human Resources and Medical Records
- System Design Driven by CSMC Clinical Systems Security Architecture
- Layered Approach: “Defense in Depth”

Defense in Depth

- Encryption
- Firewall and Server Configuration
- Authentication
- Exhaustive Logging of Accesses
- Surveillance Reports
- Ad-Hoc Audit Tools

Intranet and Internet



Palm™ VII Handheld Networking Details



Palm VII Handheld

DESX

Certicom Ellyptic Key



Palm.Net® (BellSouth) Network

DESX

Certicom Ellyptic Key



Palm.Net Proxy Server

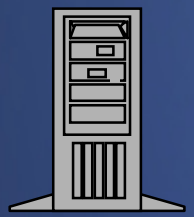
SSL HTTPS



CSMC Databases

ODBC

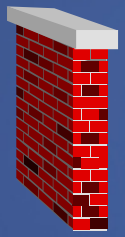
TCP/IP



CSMC Secure Web Server

SSL

HTTPS



CSMC Firewall

SSL

HTTPS



The Internet

Palm™ VII Handheld Issues

- Bandwidth 8 Kbps is comparable to slow modem
- Memory—2 MB in Palm VII, 8 MB in Palm VIIx
- No JavaScript, “cookies”, frames, nested tables
- Screen size (160x160), no horizontal scrolling
- Limited fonts, limited graphics, only 4 “colors”
- Signal is poor in some buildings and underground
- Handwritten entry of passwords
- Page cache persists across sessions
- Ongoing Palm.Net® costs and account management

Contact Information

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www.csmc.edu



Q&A


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innovate
communicate
palm
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